Ticket Booking and Reservation System

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Problem Analysis – Ticket Booking and Reservation System:

The Ticket Booking and Reservation System is designed to provide a centralized online platform for searching, reserving, booking, and managing tickets for transportation services such as buses, trains, and flights. Traditional ticket booking methods may involve manual processes, long waiting times, difficulty checking seat availability, and challenges in managing cancellations and refunds. The proposed system aims to overcome these issues through an automated and user-friendly application.

The system has four major modules: Login and User Management, Ticket Search and Reservation, Payment and Booking Management, and Admin and Schedule Management. Customers can register, log in, search for available schedules, check seat availability, select seats, reserve tickets, make payments, and receive electronic tickets. They can also view their booking history, cancel eligible bookings, and request refunds. Administrators can manage schedules, routes, ticket prices, user information, seat inventory, and booking records.

The system must ensure that each booking is associated with a customer and contains one or more tickets. Each ticket must be linked to a valid schedule and an allocated seat. Payment processing is handled through an external payment gateway, and the system must maintain accurate booking, payment, ticket, and refund statuses.

The major challenges include maintaining real-time seat availability, preventing duplicate reservations and overbooking, handling payment failures, managing reservation expiry, and ensuring data consistency during cancellations and schedule updates. Security is also essential to protect user information and restrict administrative operations to authorized users.

The proposed system addresses these problems by automating the booking workflow, providing accurate schedule and seat information, integrating secure payment processing, and maintaining centralized booking records. The overall objective is to reduce manual effort, improve booking efficiency, prevent reservation conflicts, and provide a reliable ticket booking experience for customers and administrators.

Module 1 – Login & User Management

Scope:

This module handles user authentication, registration, profile management, and role-based access to the system.

Objectives:

- Allow users to register and log in securely.
- Authenticate users based on their roles.
- Manage user profiles and passwords.
- Ensure secure access to system functionalities.

Challenges:

- Secure password storage and authentication.
- Prevent unauthorized access.
- Handle forgotten password requests.
- Manage multiple user roles (Customer and Admin).

Module 2 – Ticket Search & Reservation

Scope:

This module enables customers to search available tickets, view schedules, check seat availability, select seats, and reserve tickets before payment.

Objectives:

- Provide quick ticket search.
- Display accurate schedules and seat availability.
- Prevent double booking of seats.
- Create temporary reservations before payment.

Challenges:

- Maintaining real-time seat availability.
- Handling simultaneous reservations.
- Preventing duplicate bookings.
- Managing reservation expiry if payment is not completed.

Module 3 – Payment & Booking Management

Scope:

This module manages payment processing, booking confirmation, ticket generation, cancellation, and refunds.

Objectives:

- Secure payment processing.
- Generate electronic tickets.
- Maintain booking history.
- Process cancellations and refunds efficiently.

Challenges:

- Handling payment failures.
- Integrating with external payment gateways.
- Updating booking status correctly.
- Managing refund requests.

Module 4 – Admin & Schedule Management

Scope:

This module allows administrators to manage schedules, routes, pricing, users, seat inventory, bookings, and reports.

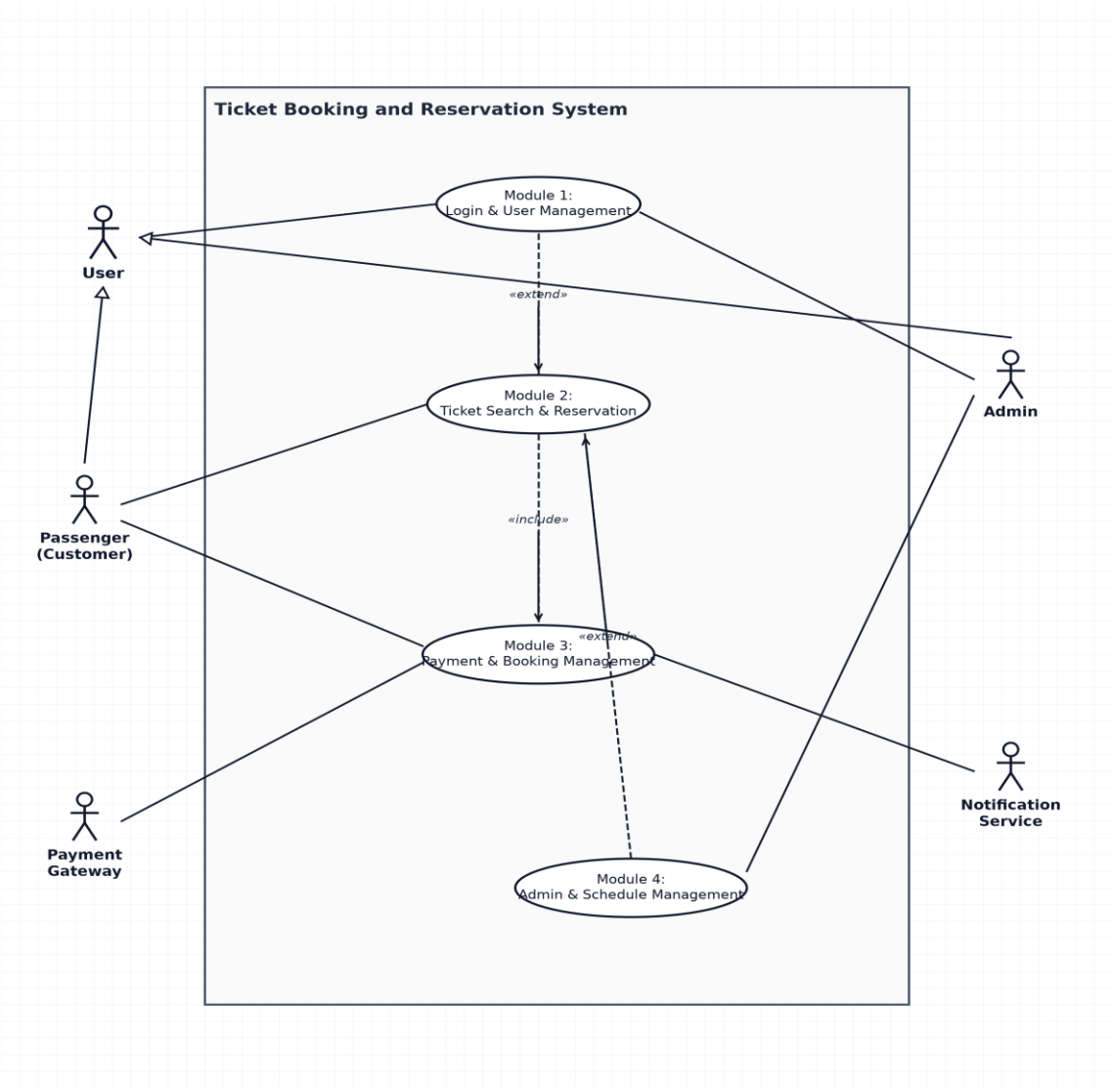
Objectives:

- Maintain accurate schedules.
- Manage ticket pricing.
- Monitor bookings.
- Generate reports for system analysis.

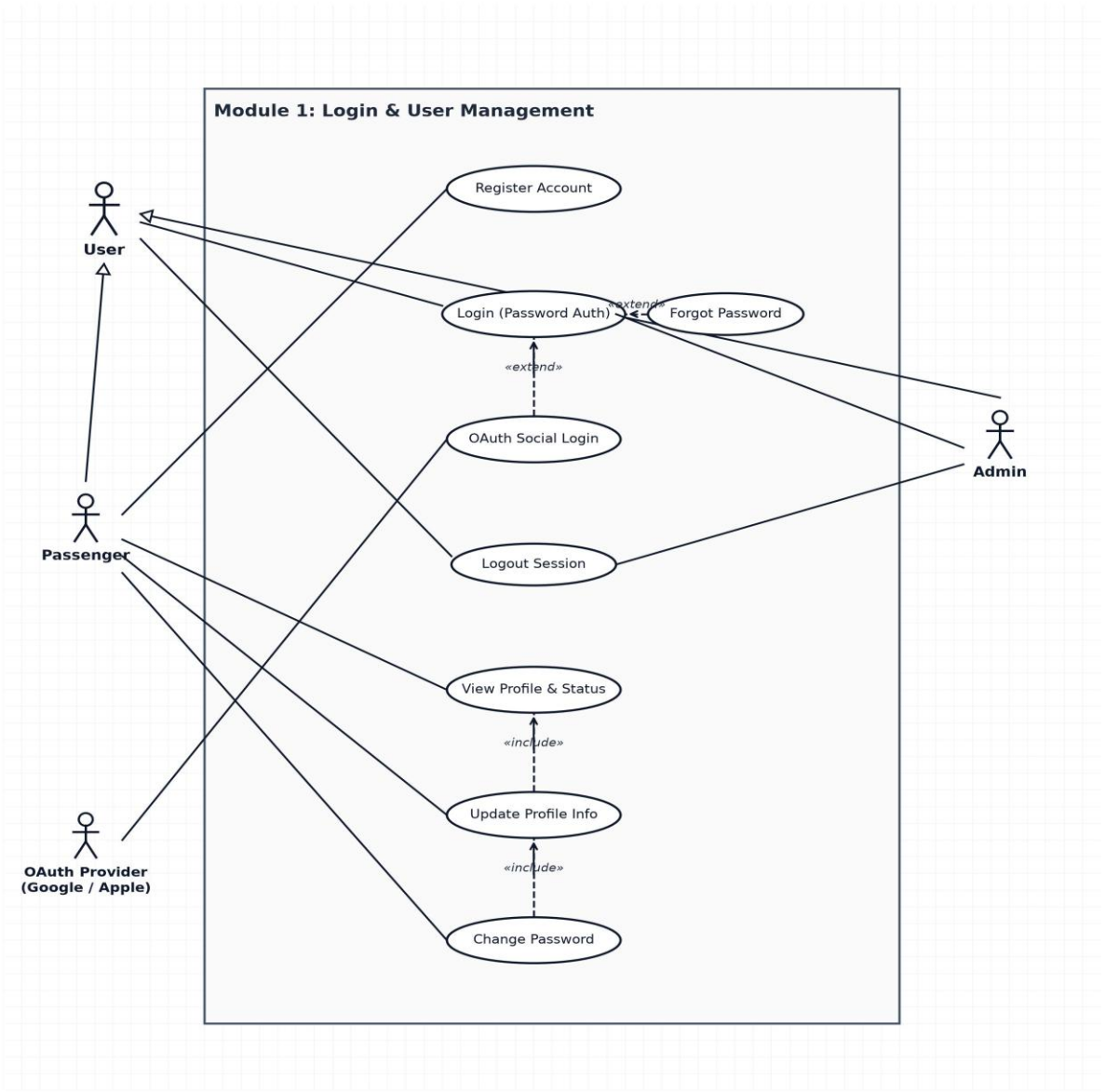
Challenges:

- Updating schedules without affecting existing bookings.
- Managing large volumes of booking data.
- Maintaining data consistency.
- Providing secure administrative access

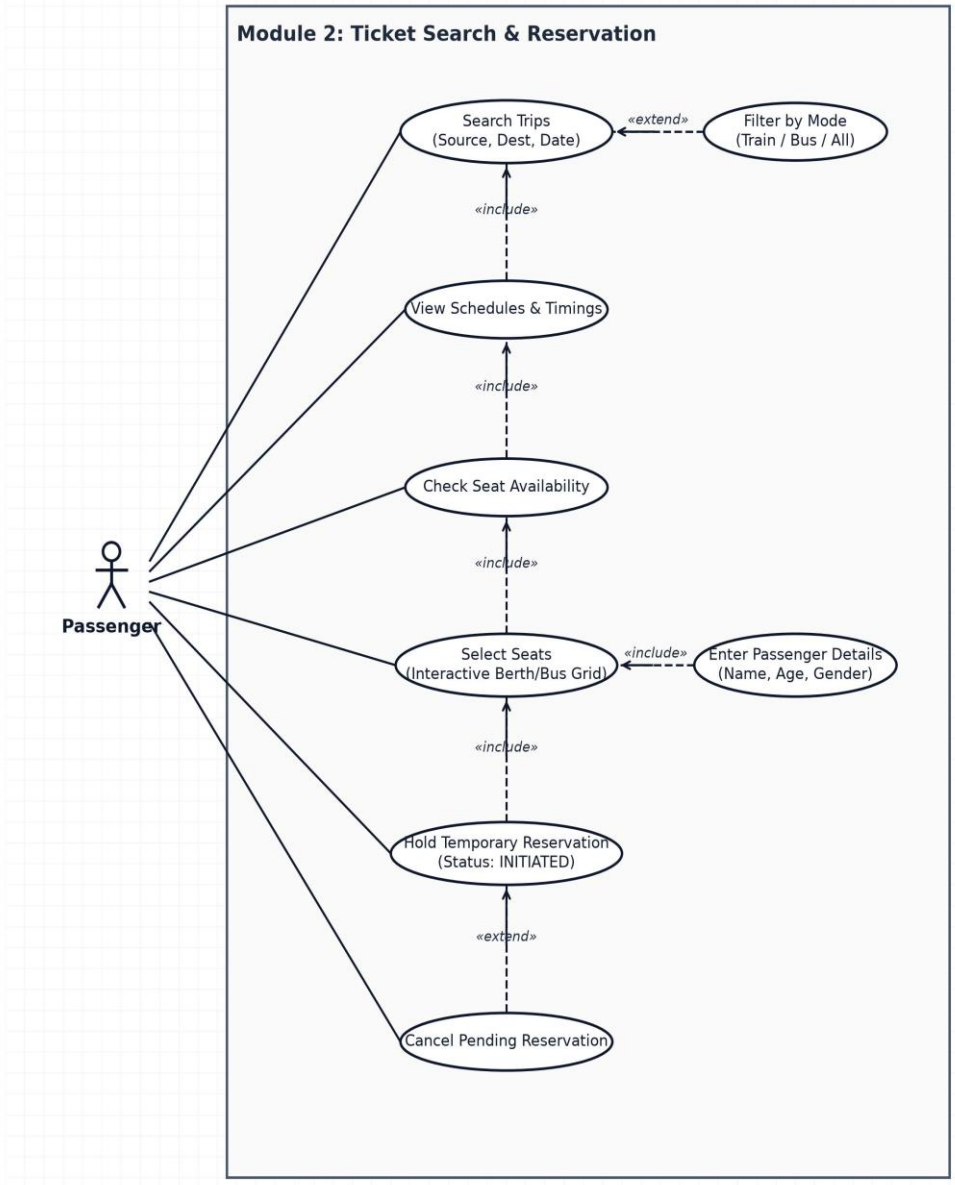
Overall Use Case Diagram



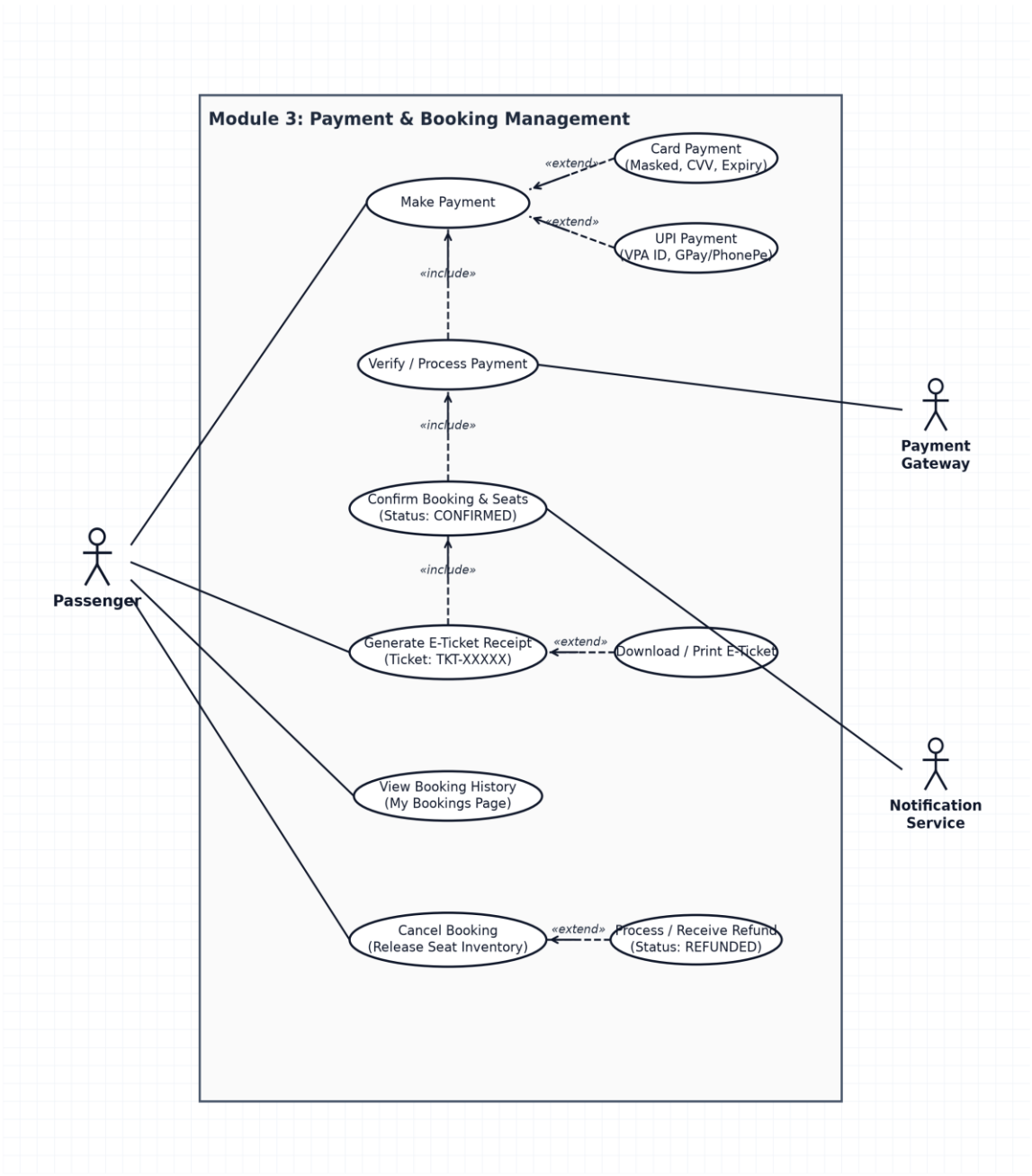
Module 1 – Login & User Management:



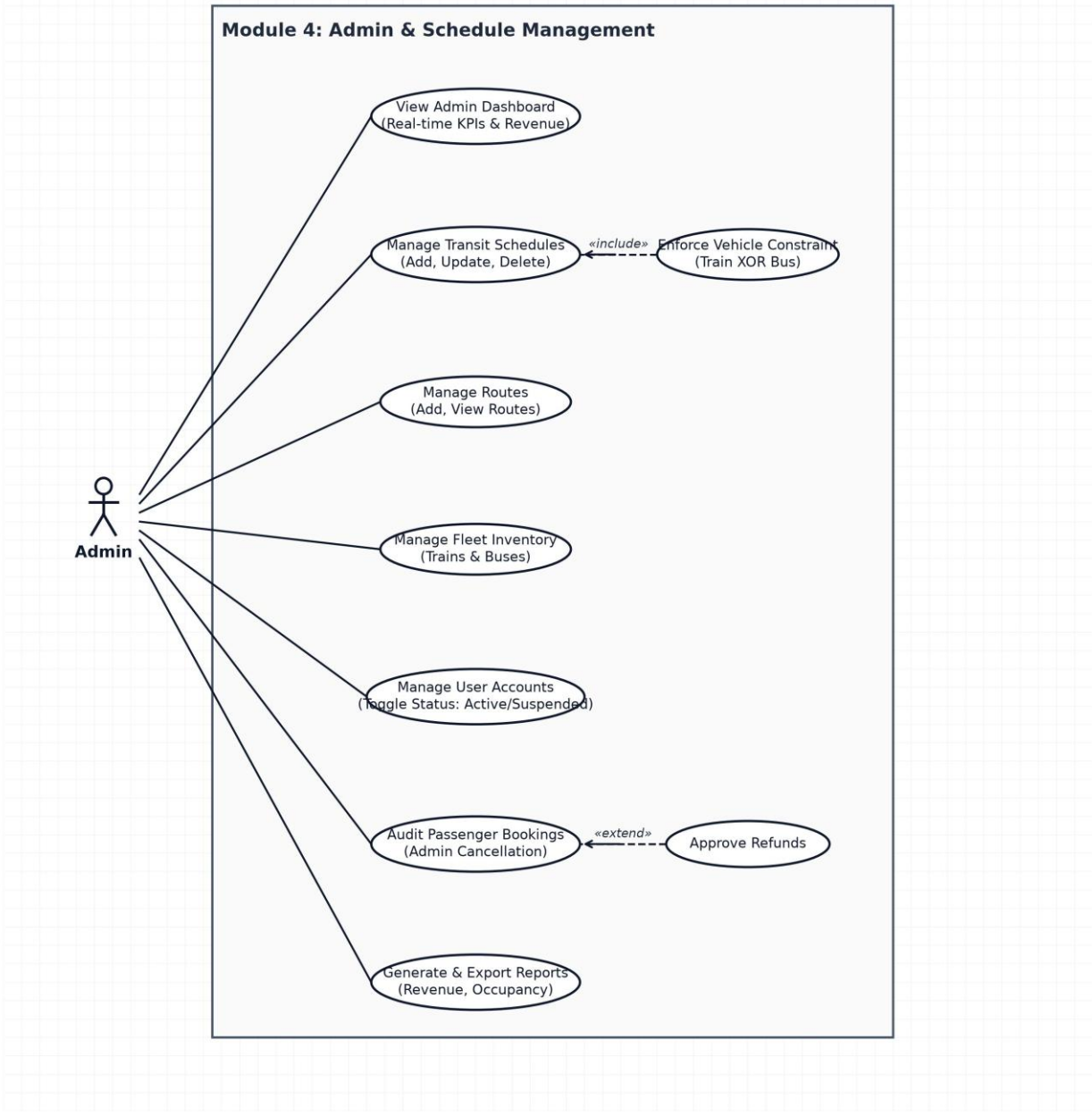
Module 2 – Ticket Search & Reservation



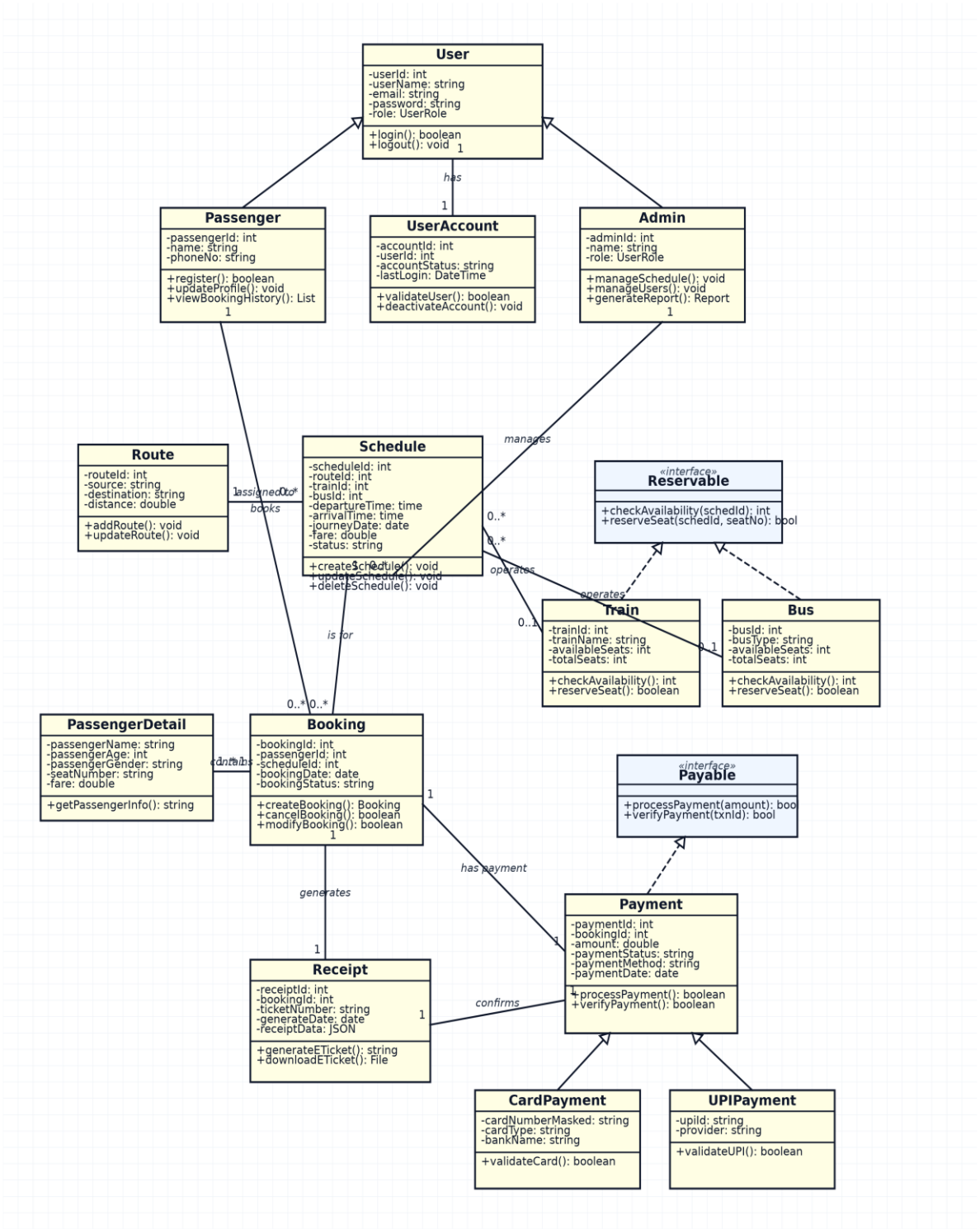
Module 3 – Payment & Booking Management



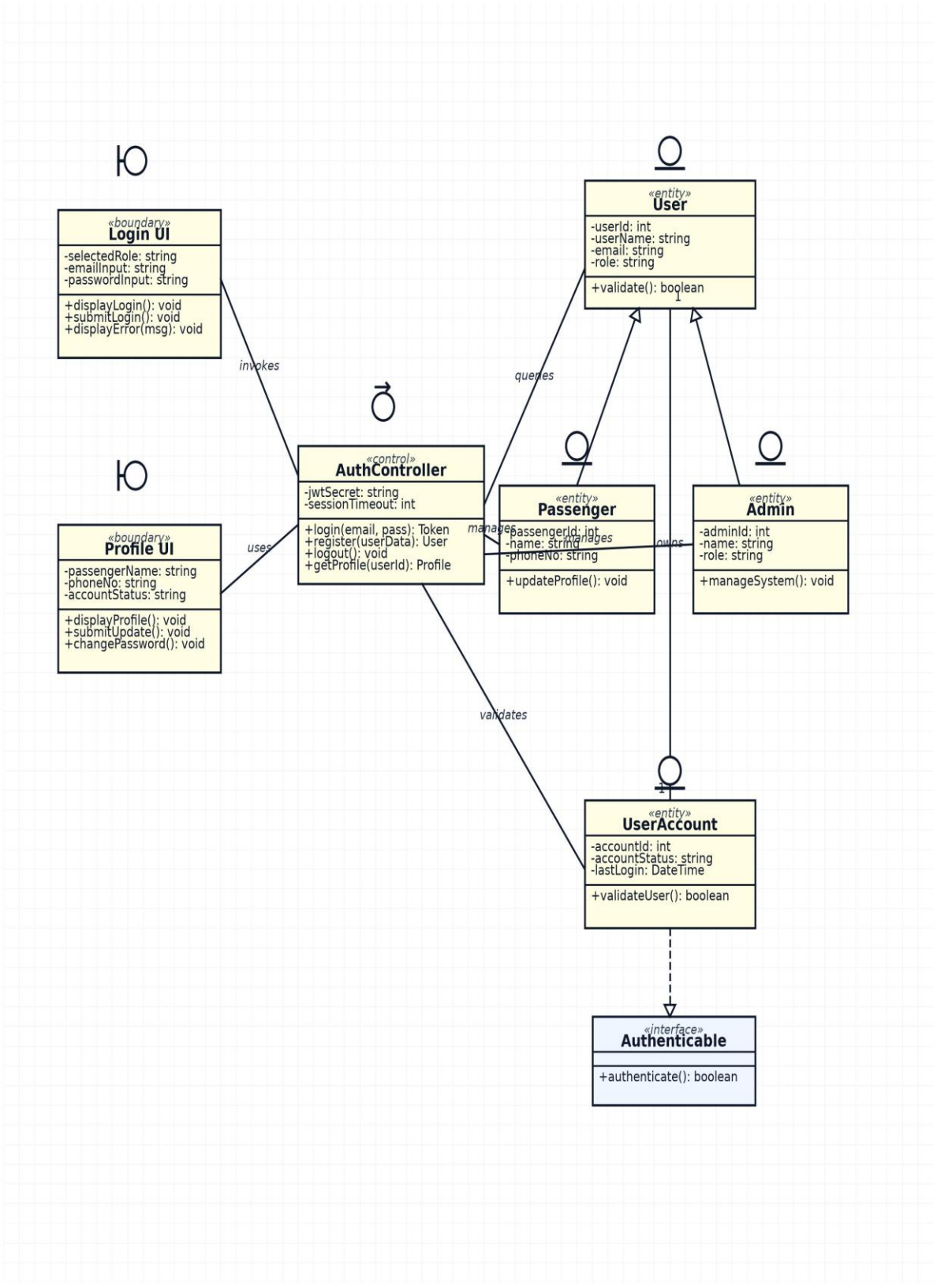
Module 4 – Admin & Schedule Management



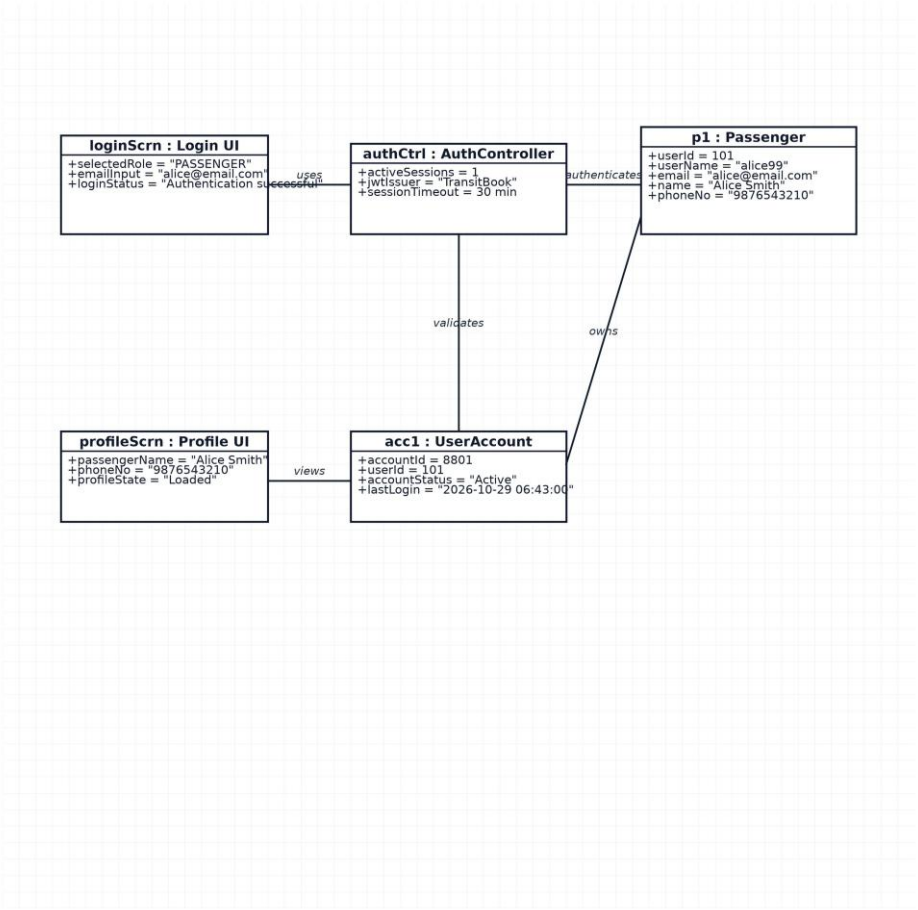
Overall Class Diagram:



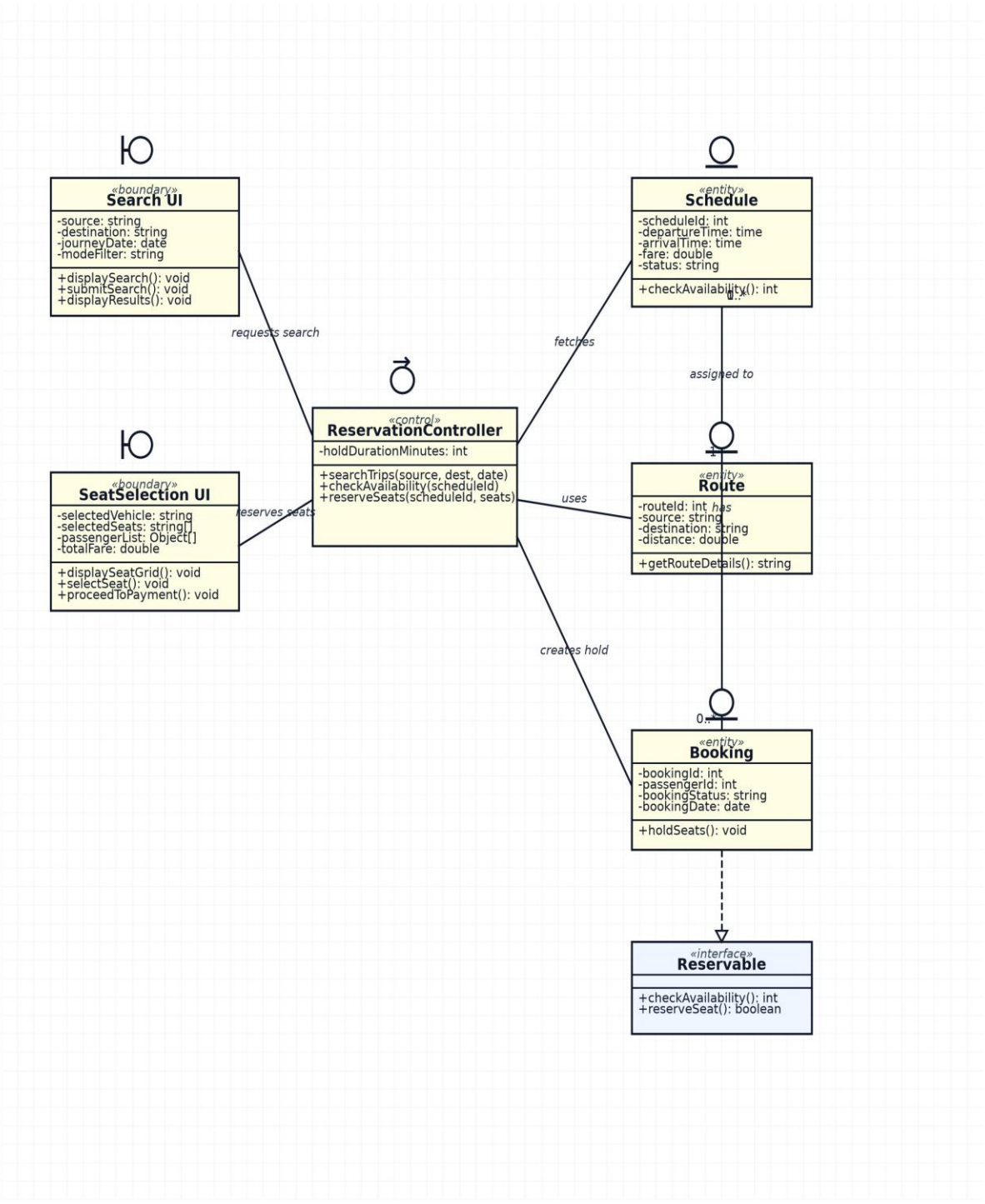
Module 1 – Login & User Management



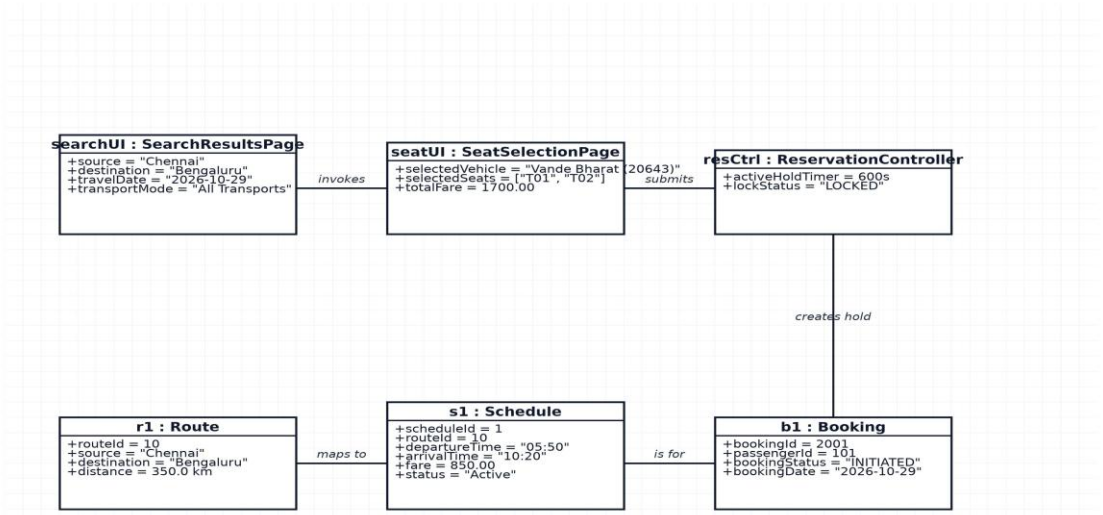
Object Diagram



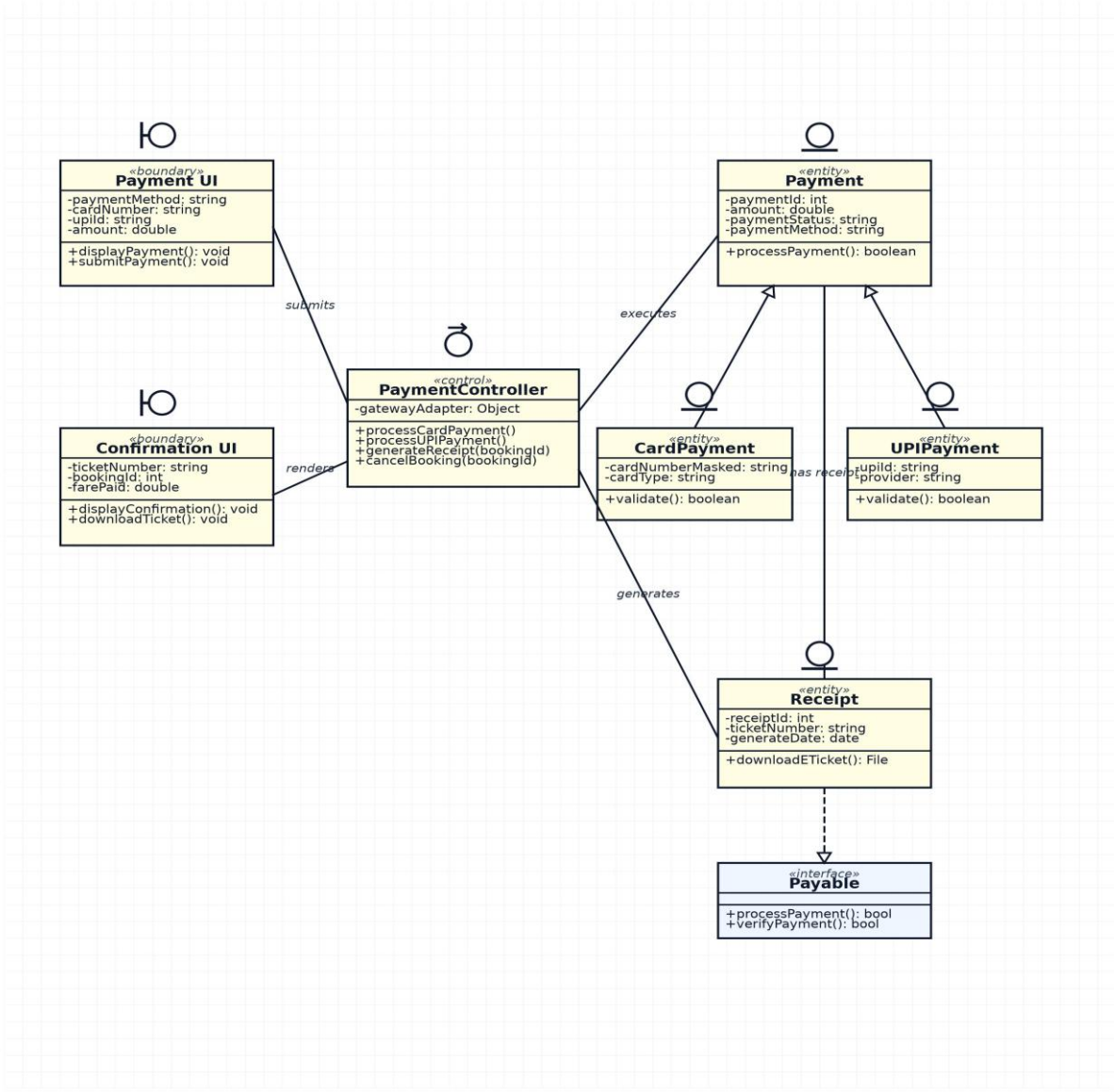
Module 2 – Ticket Search & Reservation



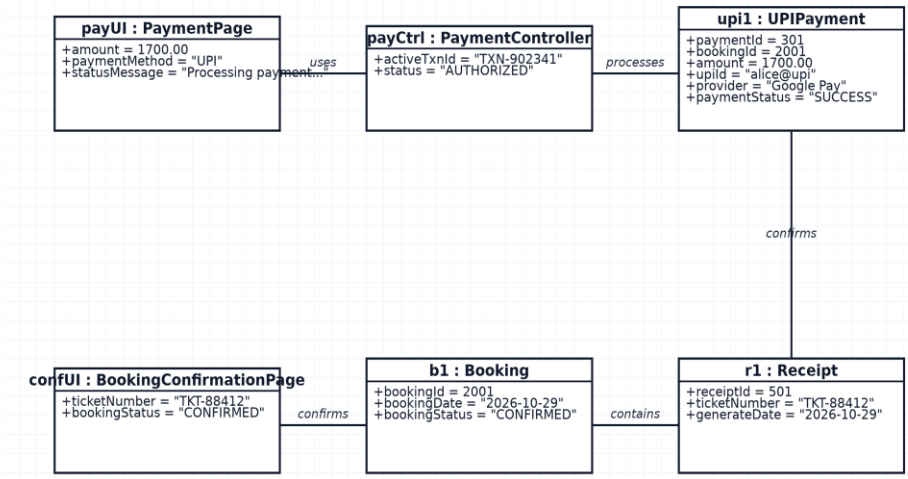
Object Diagram



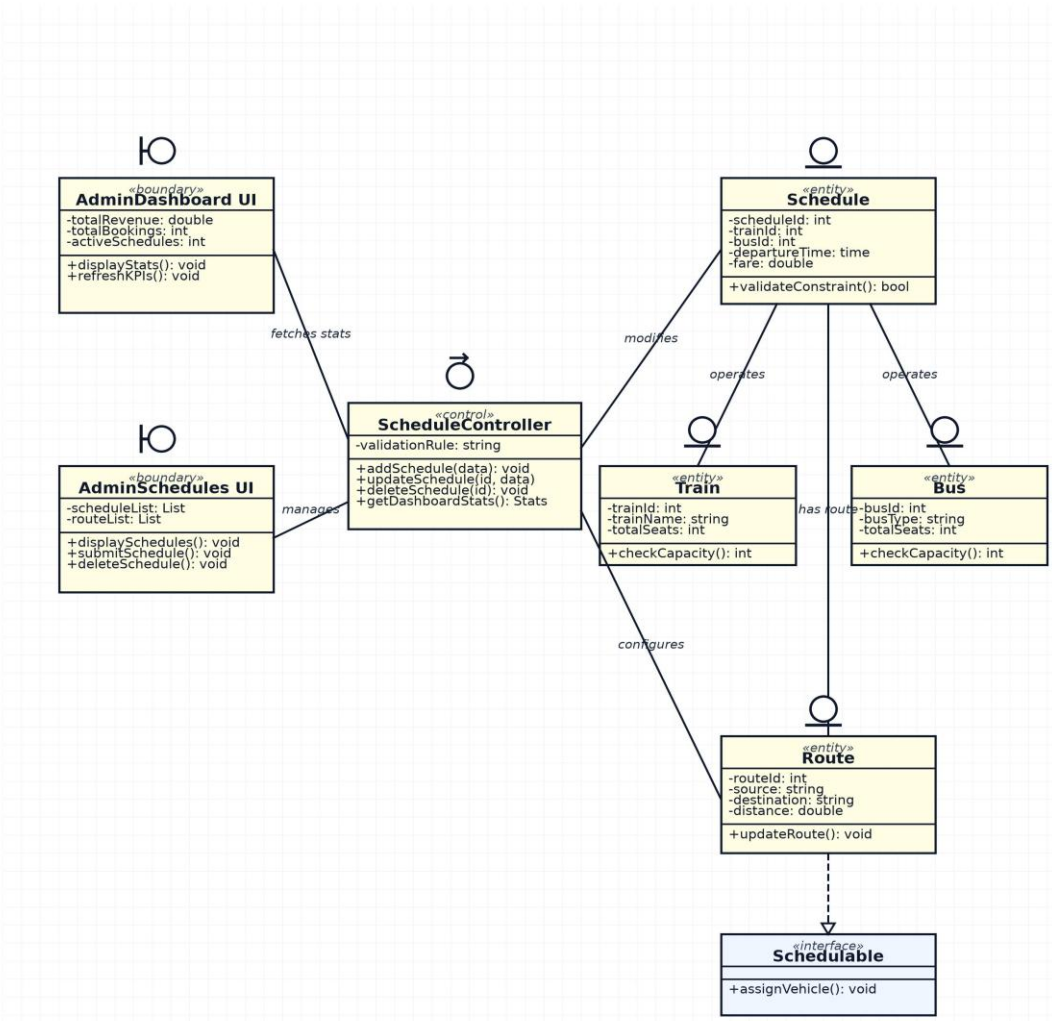
Module 3 – Payment & Booking Management



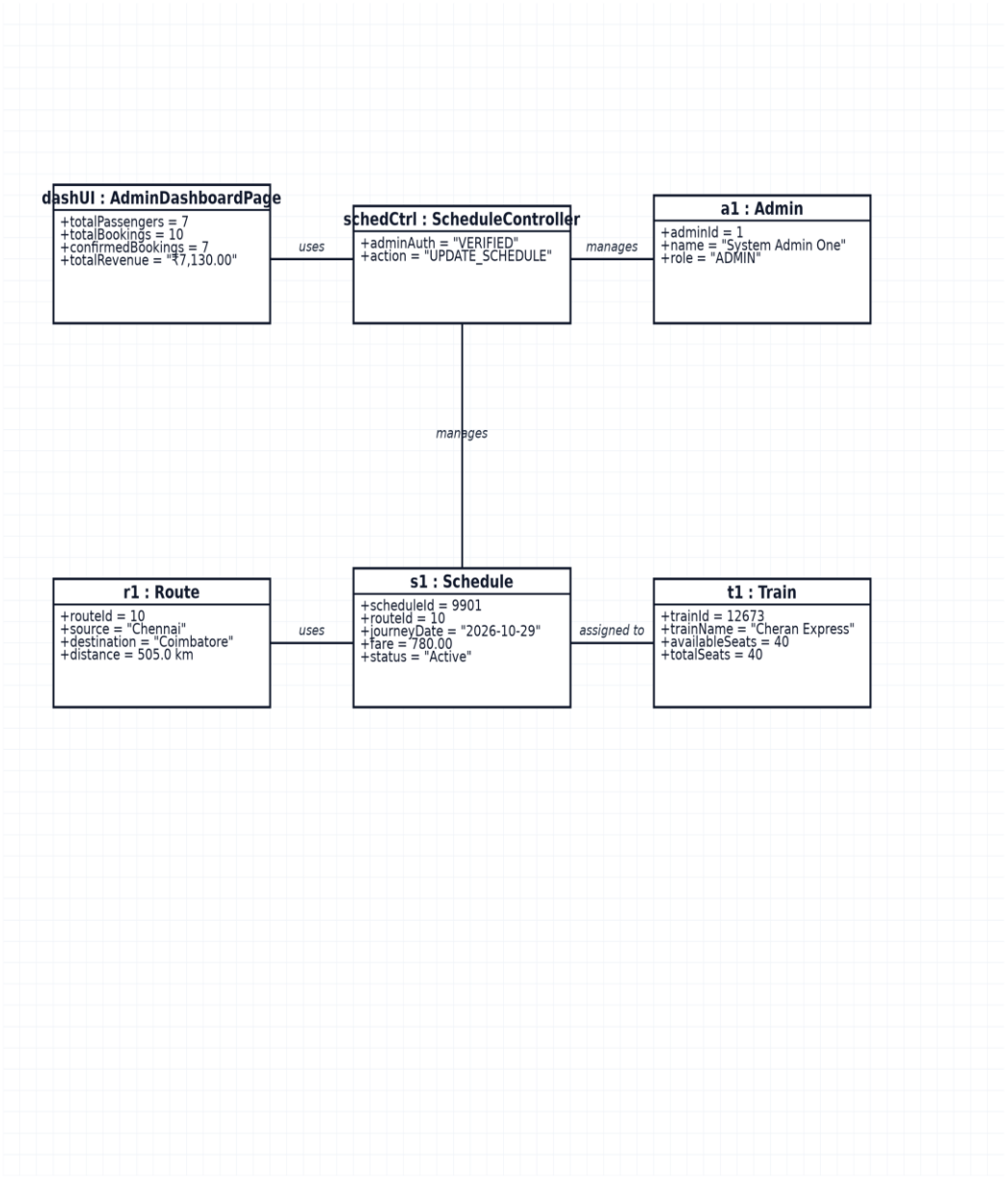
Object diagram



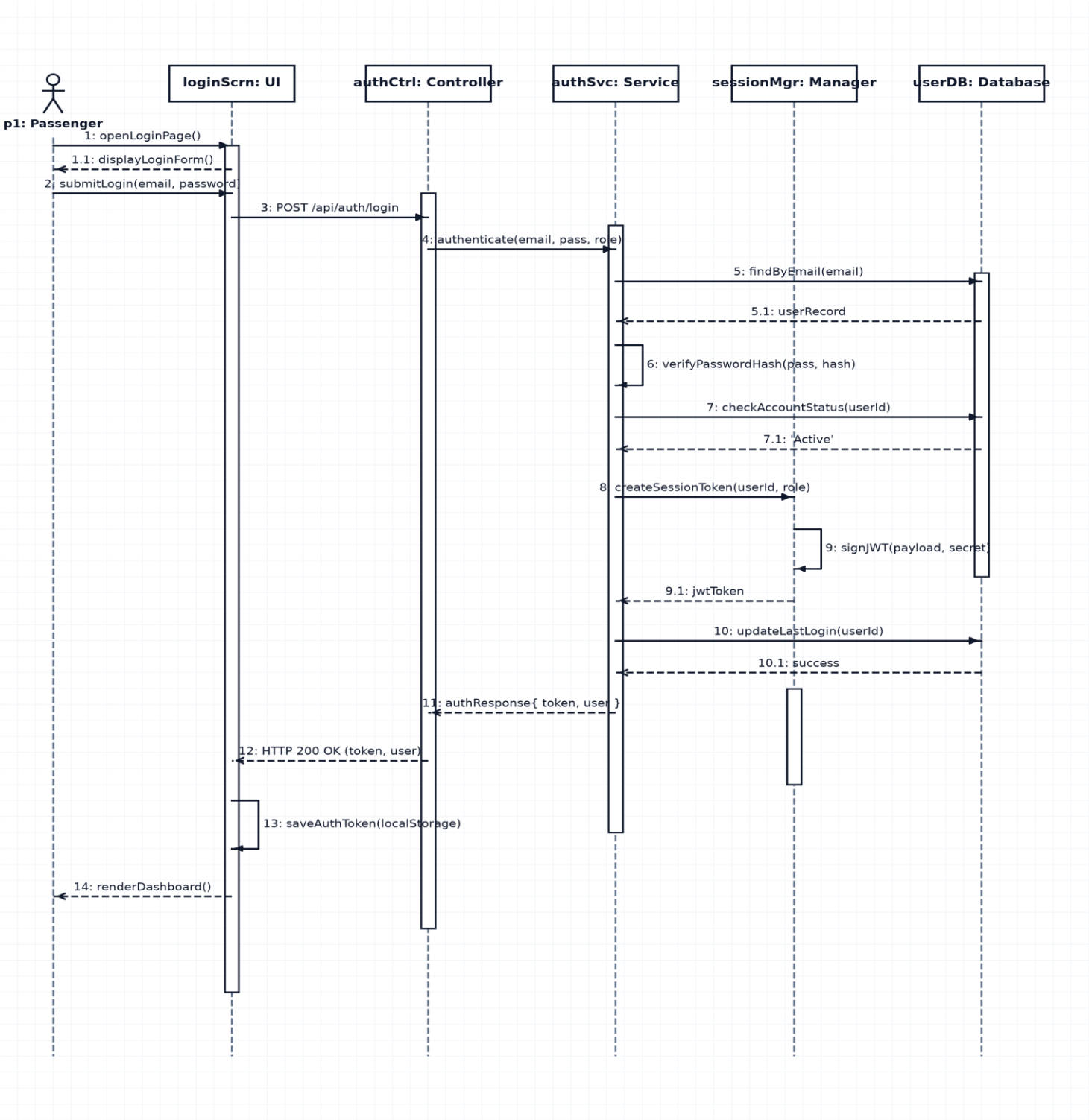
Module 4 – Admin & Schedule Management



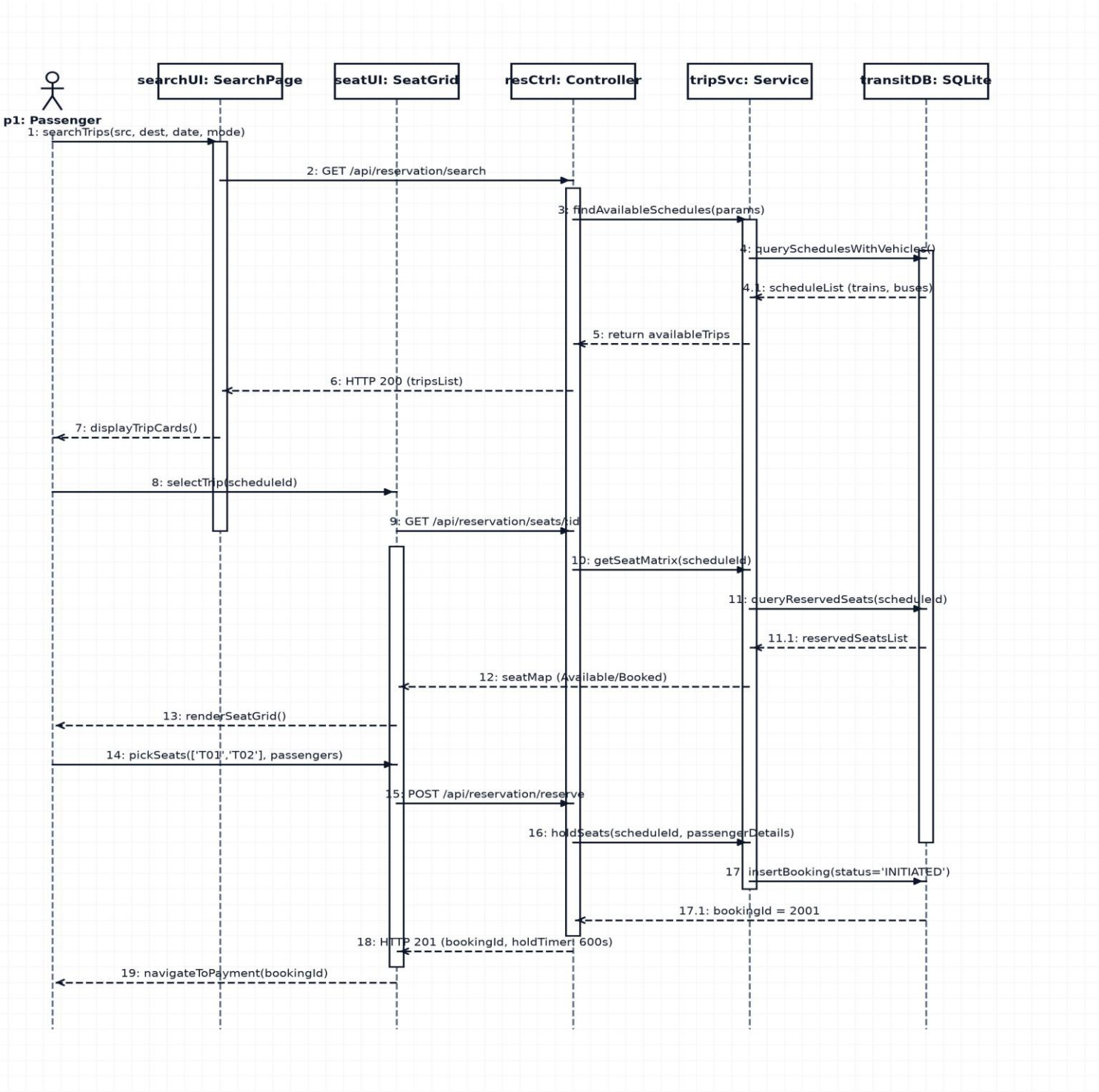
Object Diagram



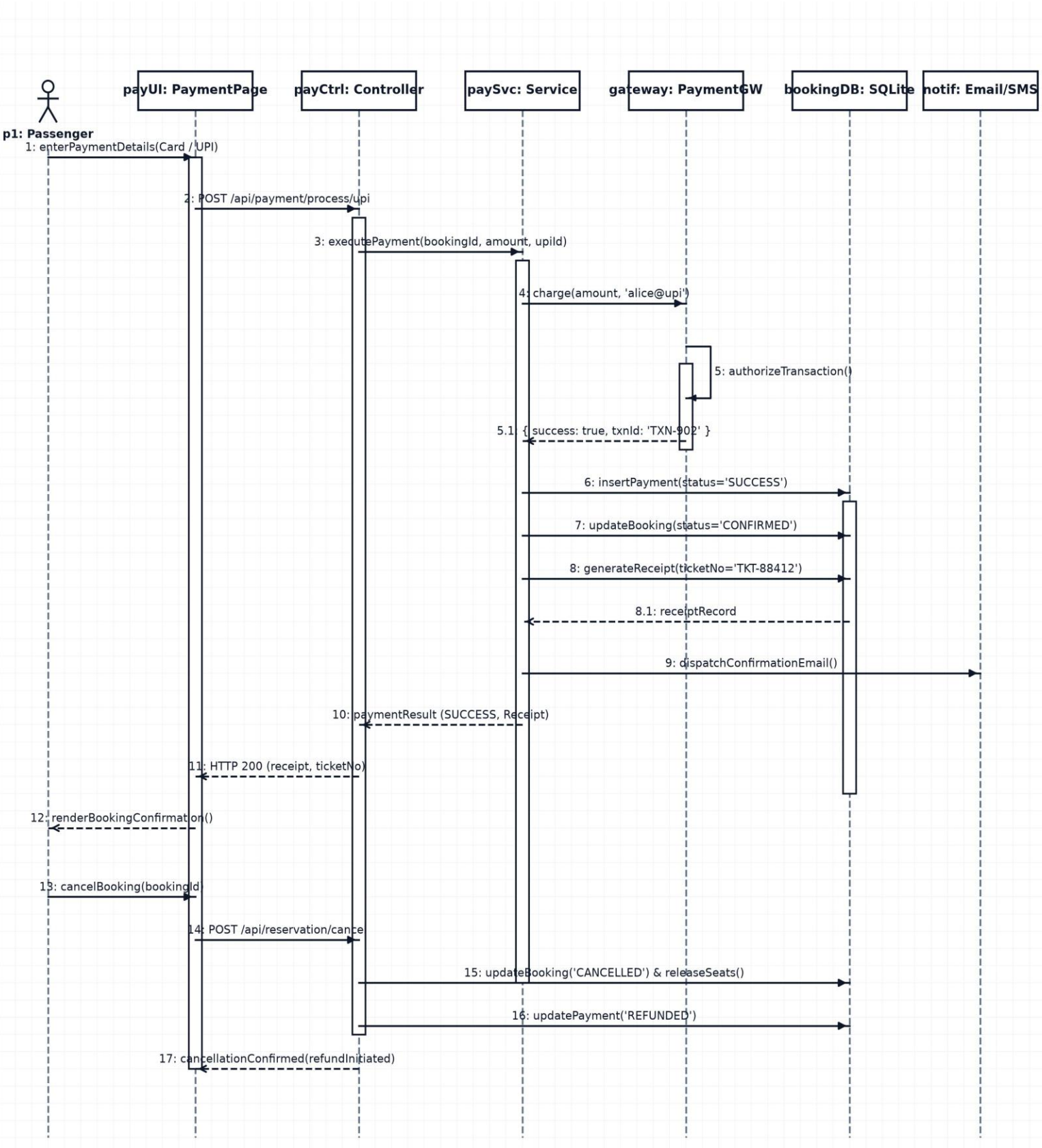
Module 1 – Login & User Management



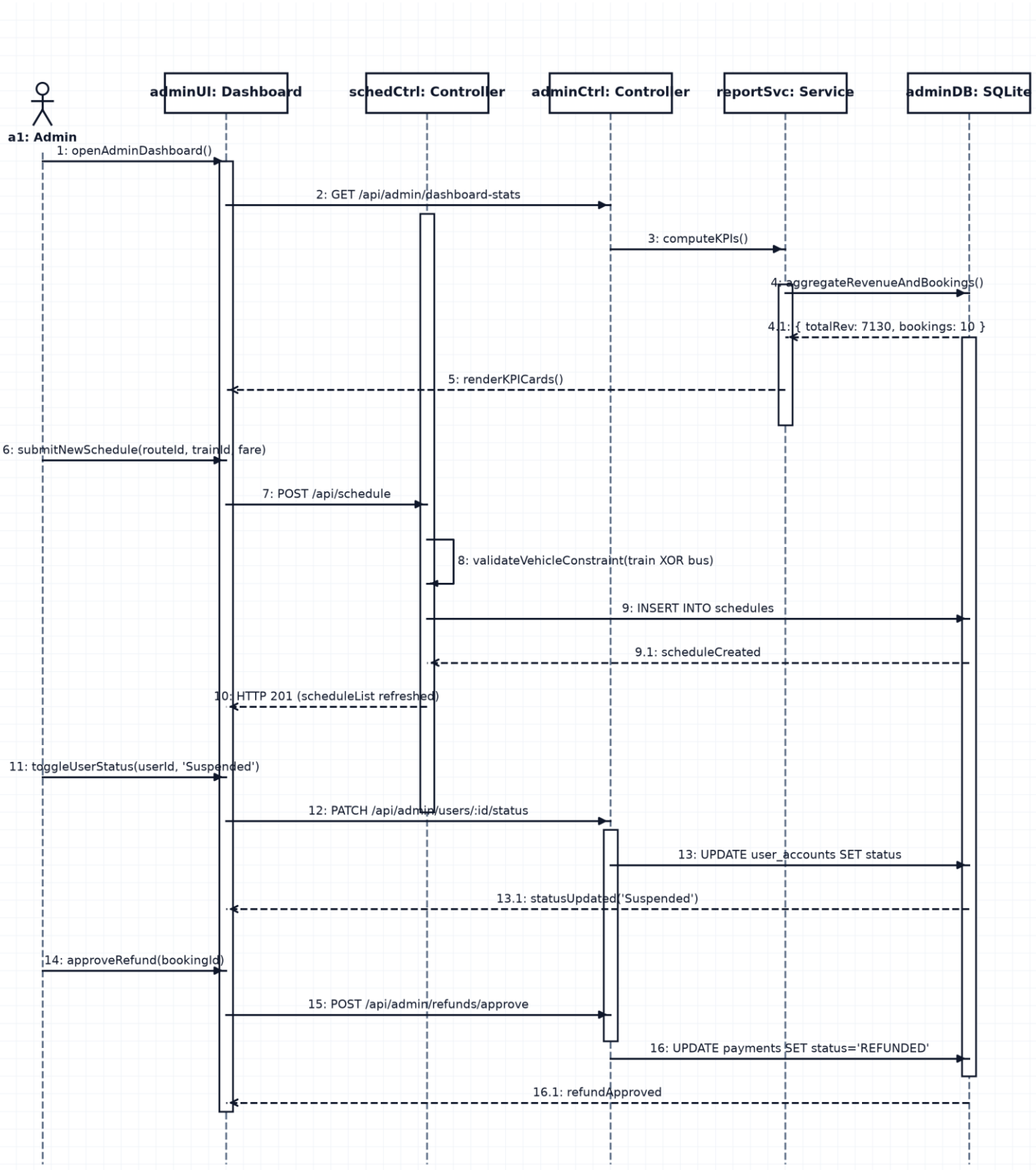
Module 2 – Ticket Search & Reservation



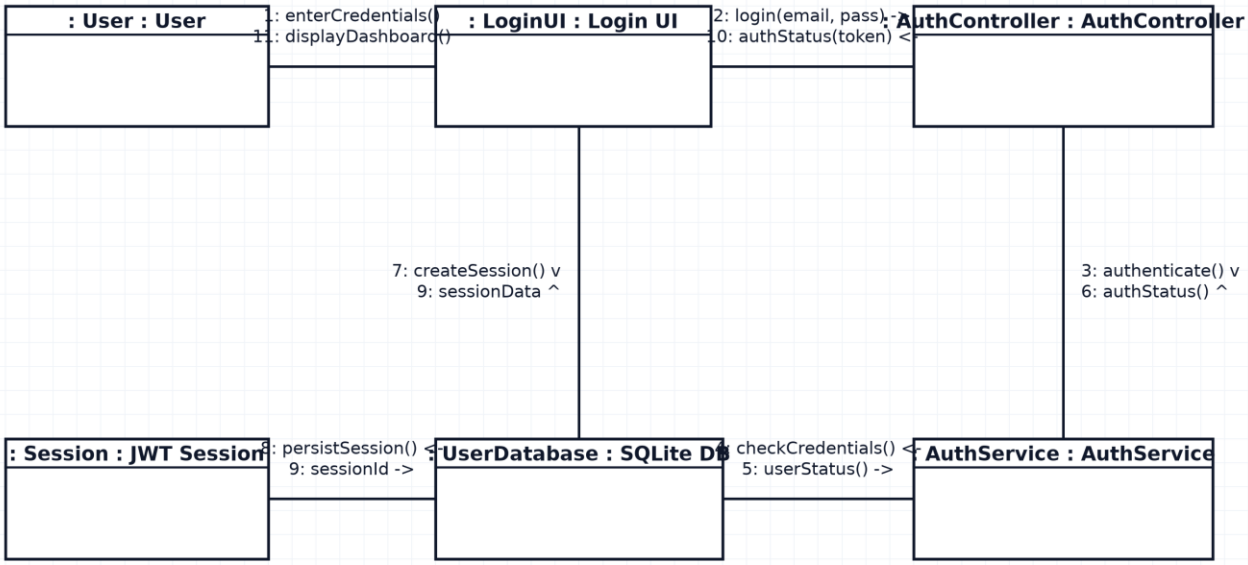
Module 3 – Payment & Booking Management



Module 4 – Admin & Schedule Management

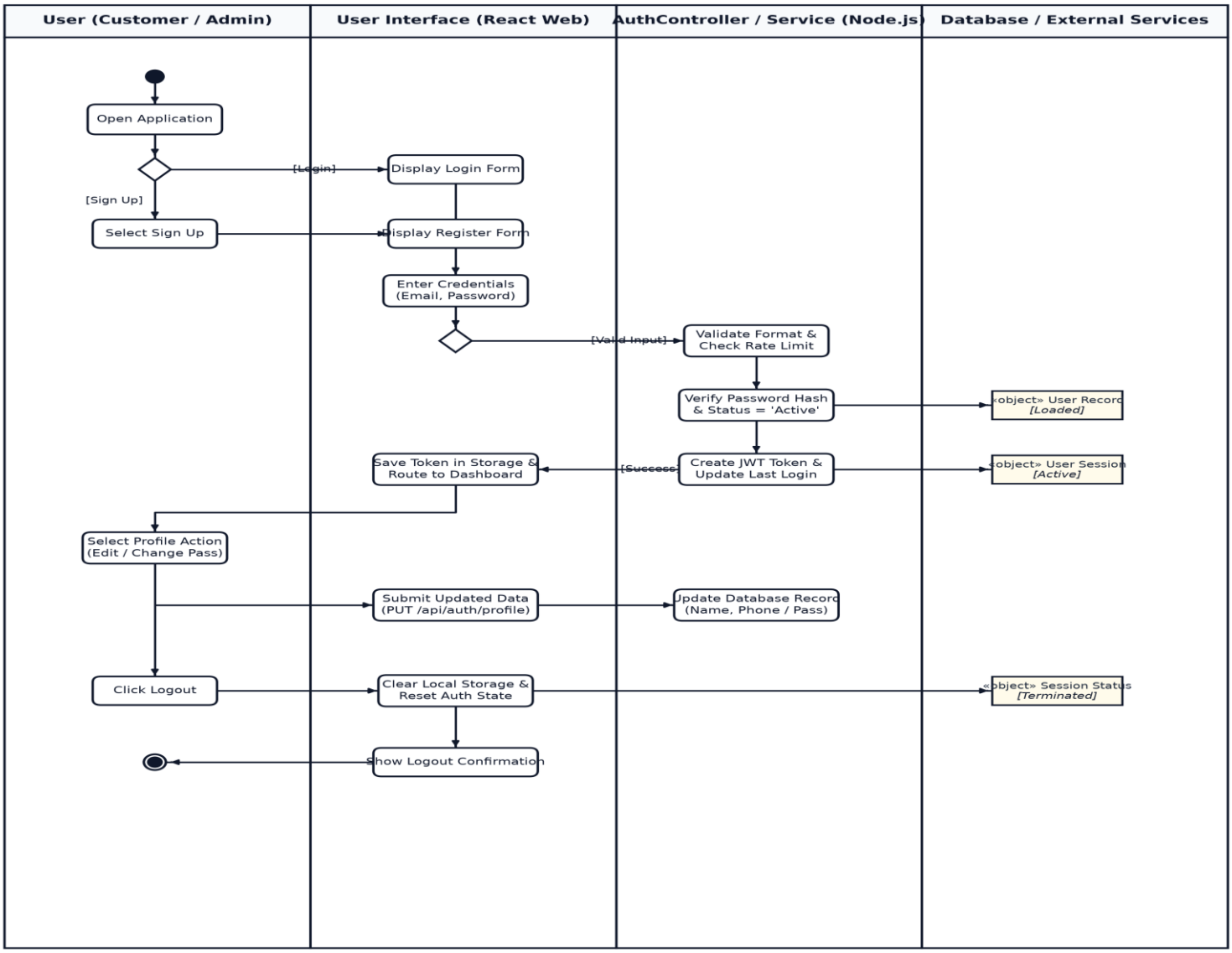


Collaboration Diagram for Login & User Management module:

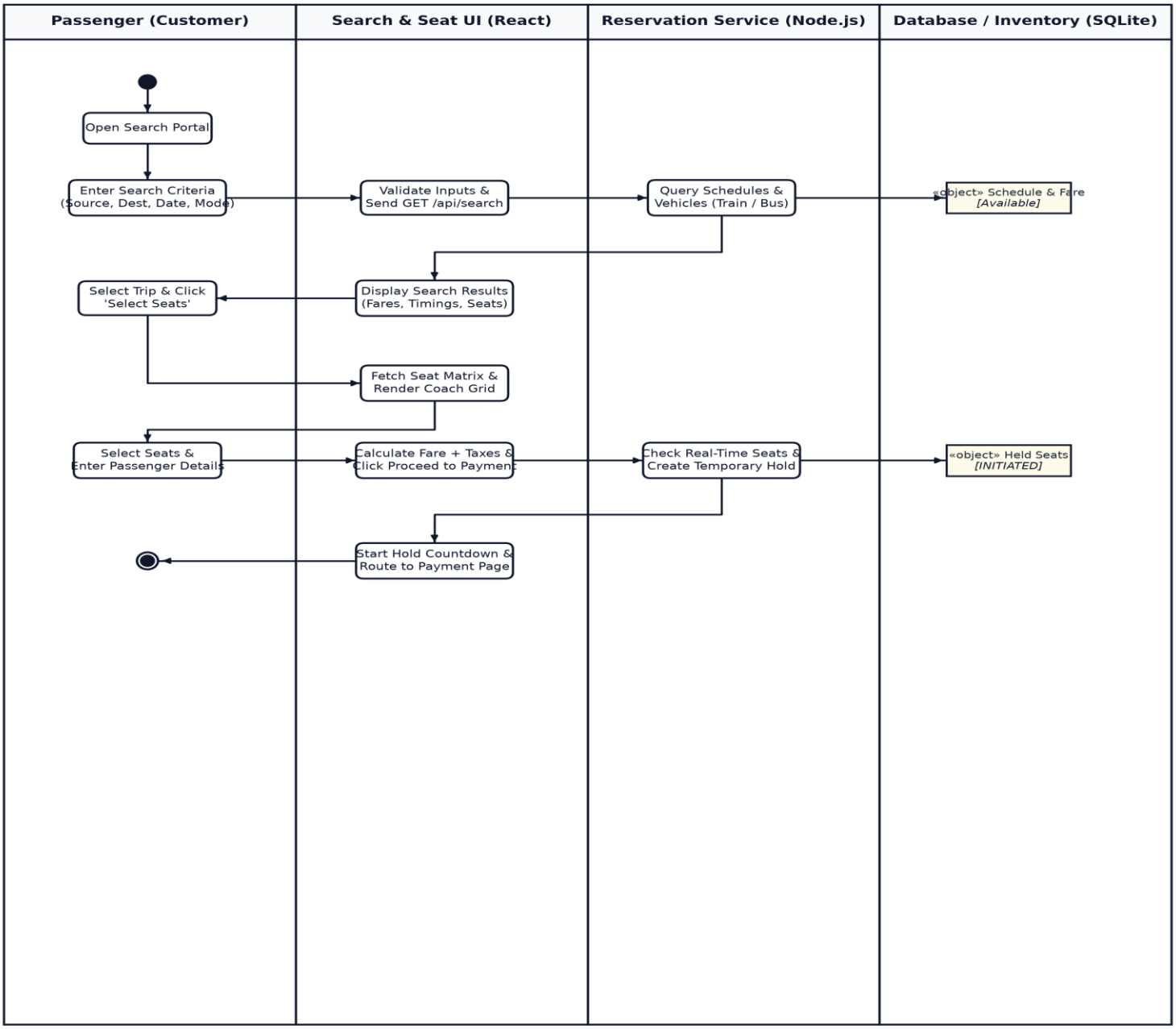


Activity Diagram

Module 1 – Login & User Management

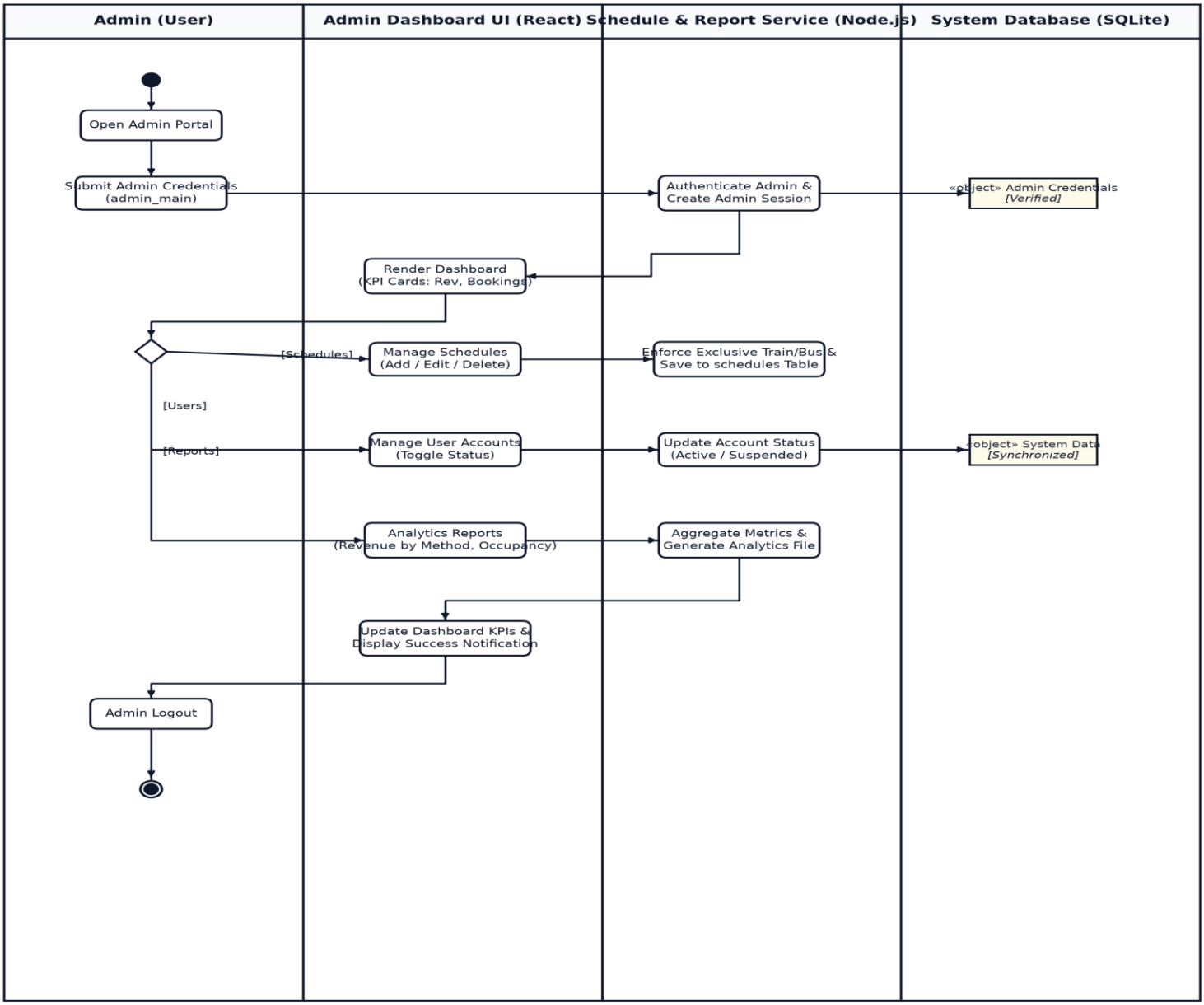


Module 2- Ticket Search & Reservation



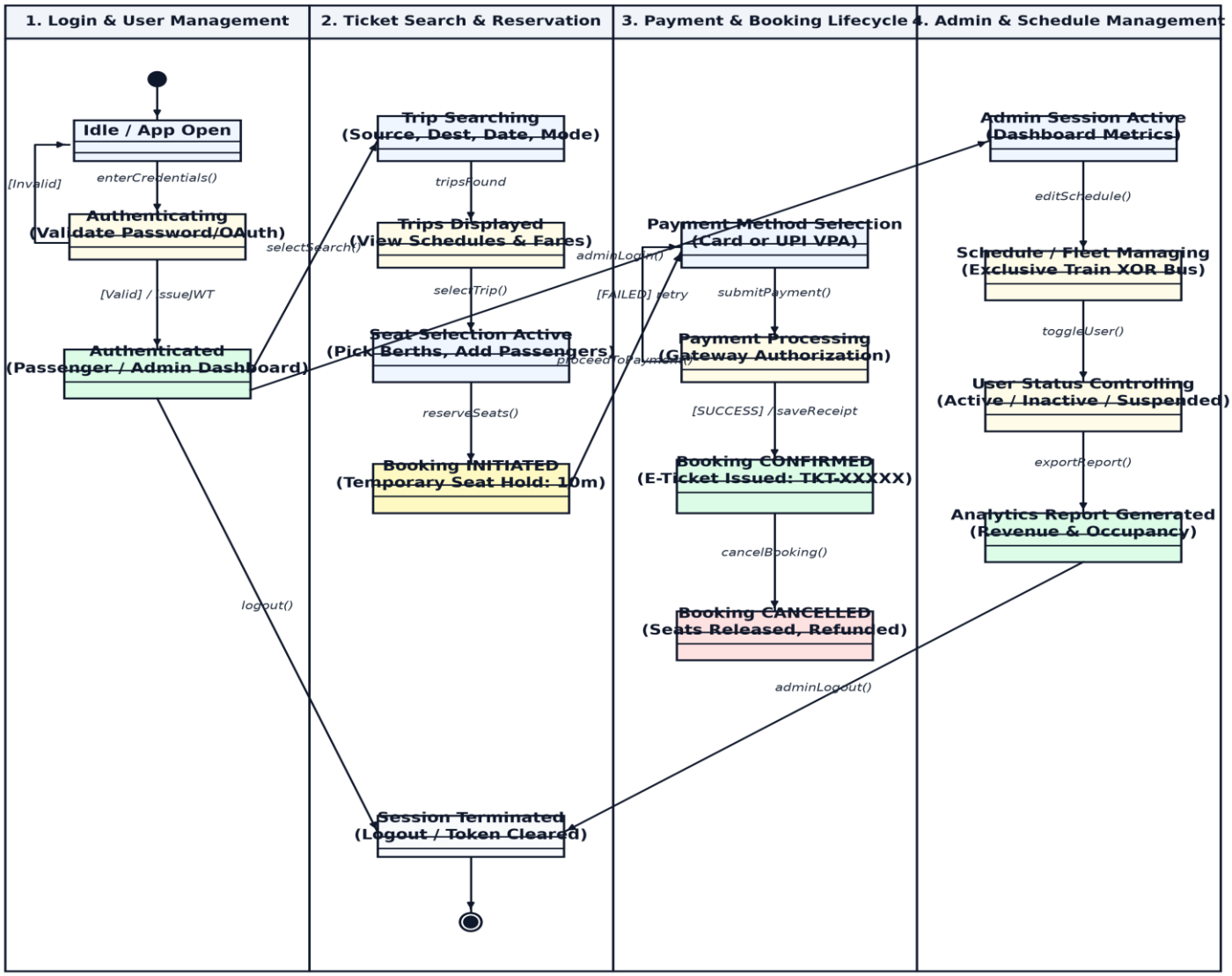
Module 3- Payment & Booking Management

Module 4- Admin & Schedule Management



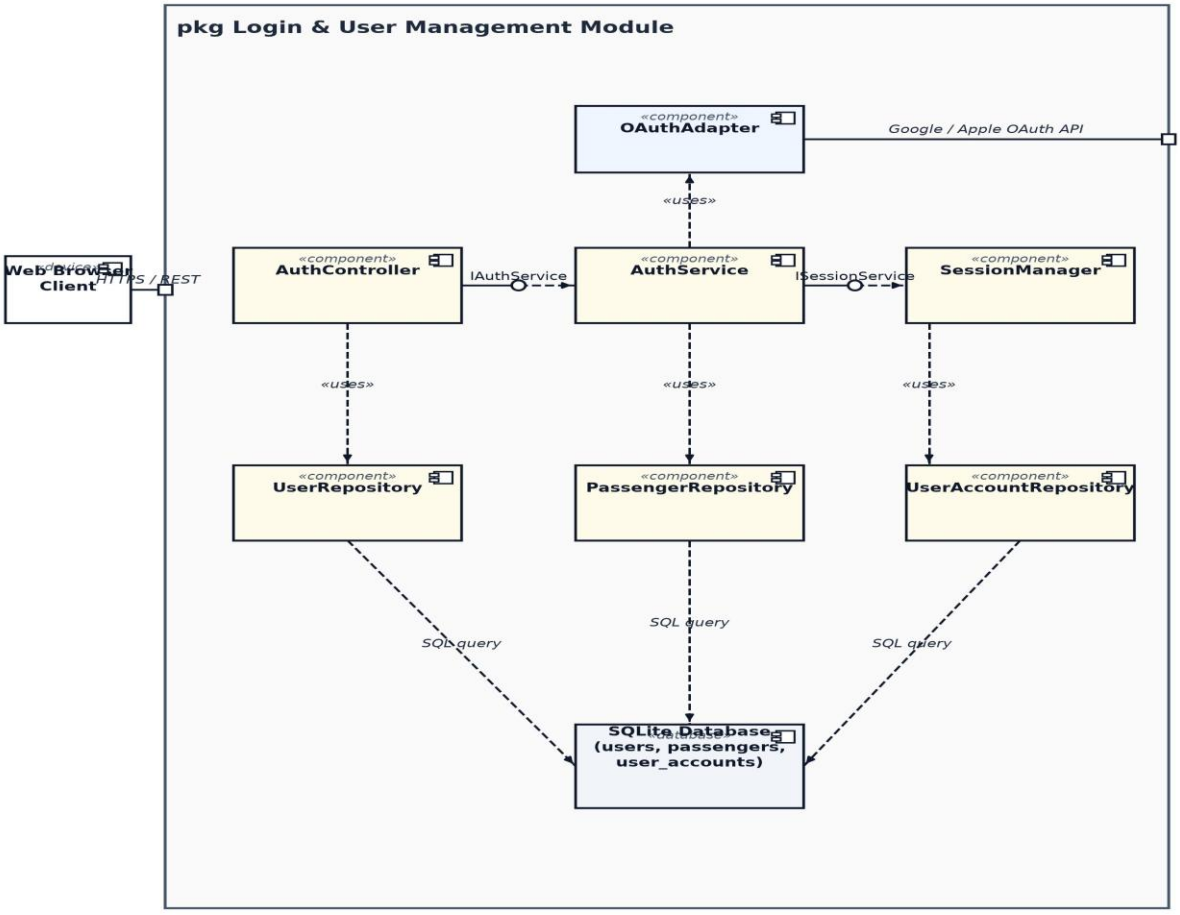
Overall Statechart diagram

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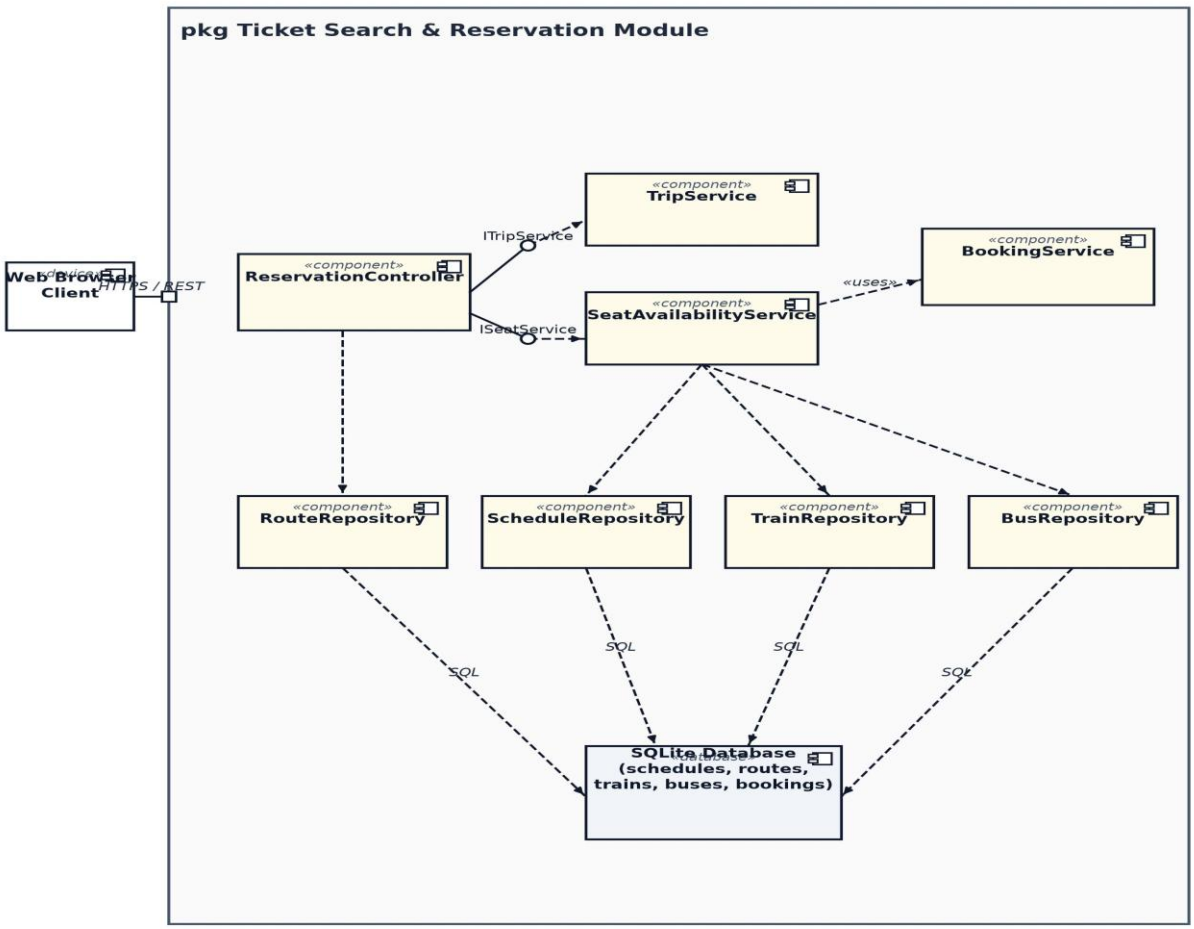


Component diagram

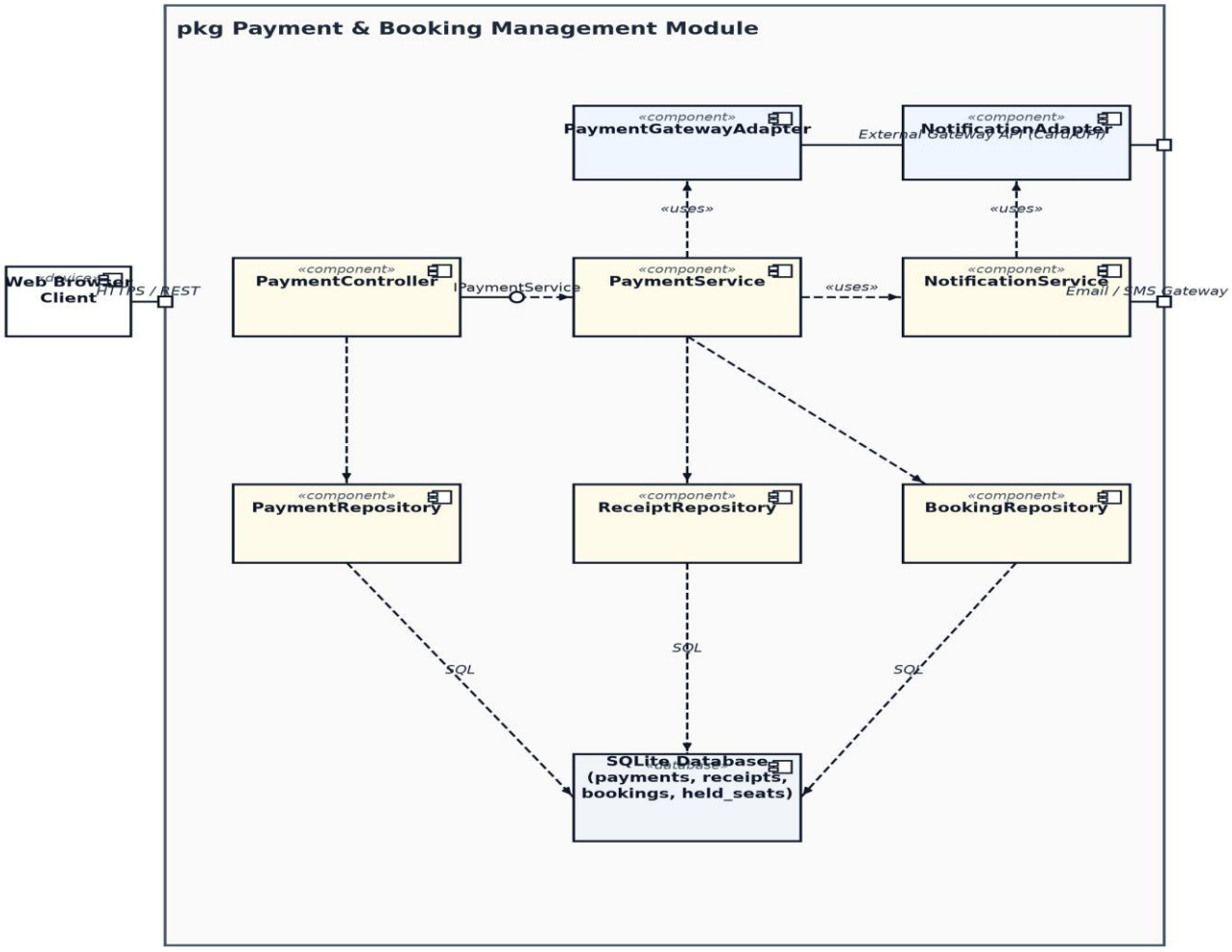
Module 1 – Login & User Management:



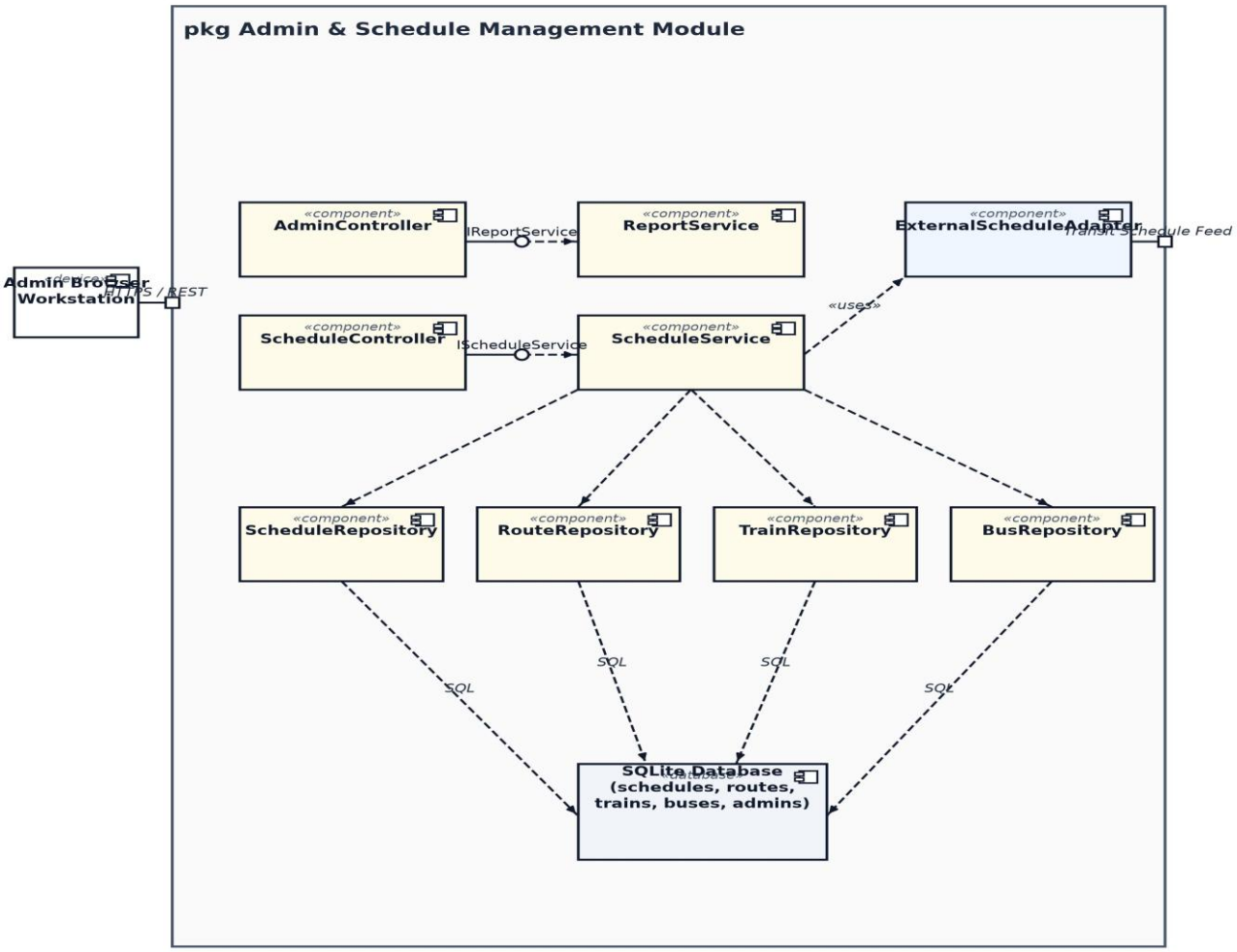
Module 2 – Ticket Search & Reservation



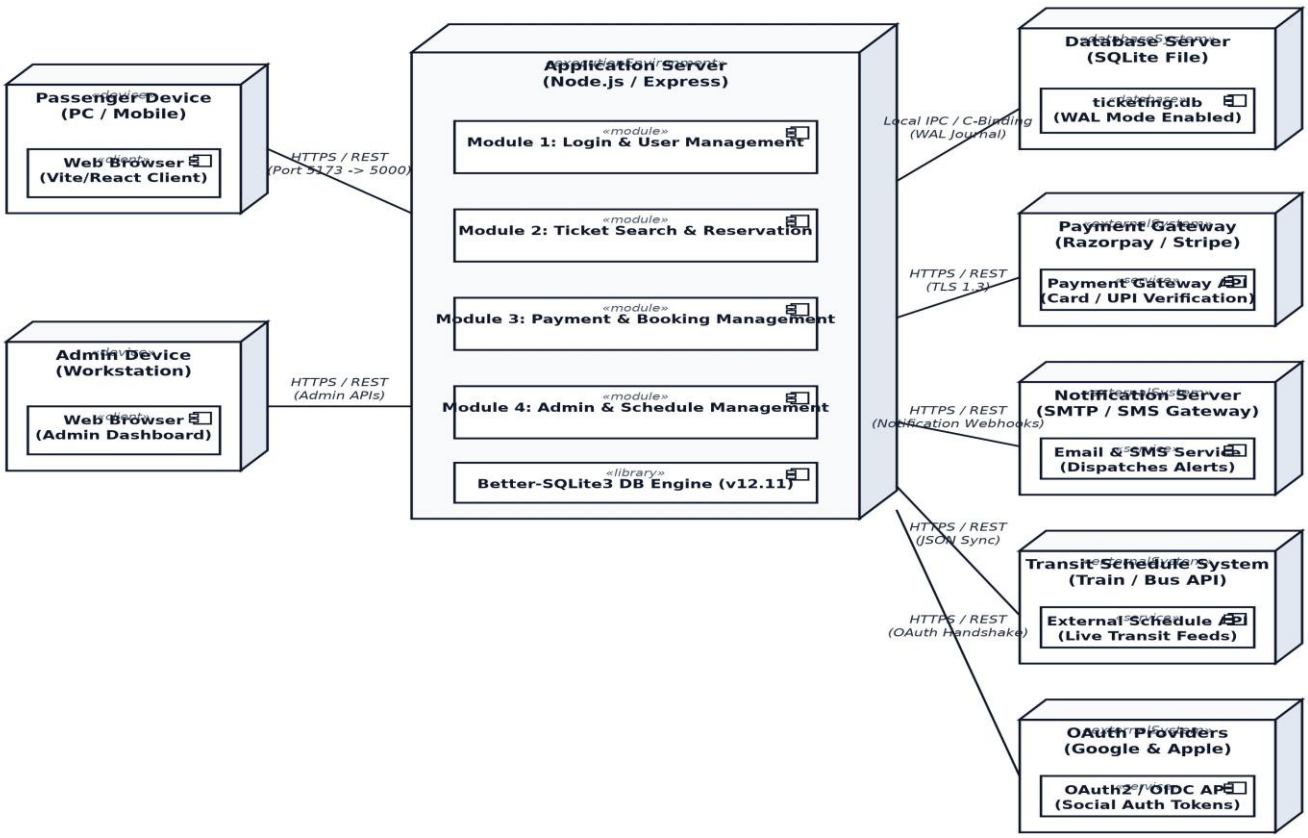
Module 3 – Payment & Booking Management



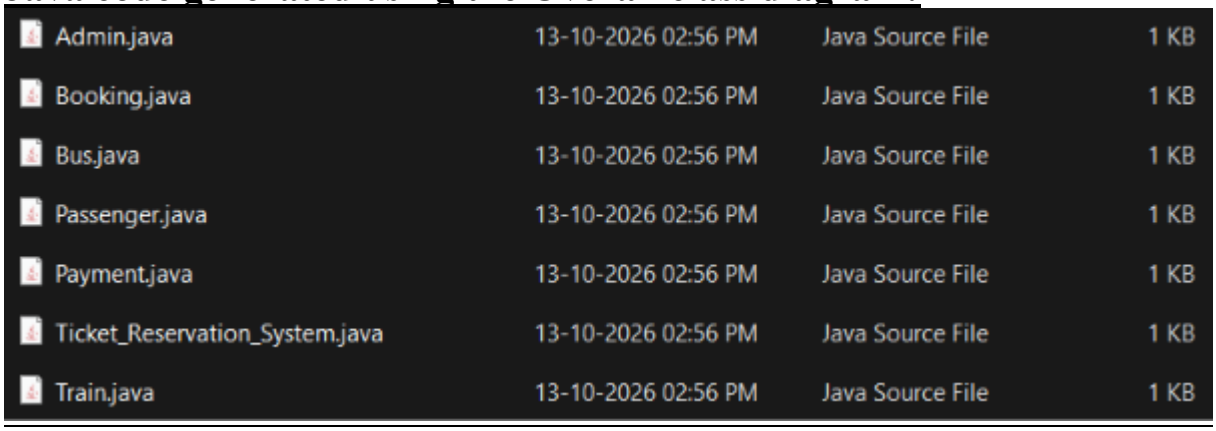
Module 4 – Admin & Schedule Management



Overall Deployment Diagram



Java code generated using the Overall class diagram:



Admin.java	13-10-2026 02:56 PM	Java Source File	1 KB
Booking.java	13-10-2026 02:56 PM	Java Source File	1 KB
Bus.java	13-10-2026 02:56 PM	Java Source File	1 KB
Passenger.java	13-10-2026 02:56 PM	Java Source File	1 KB
Payment.java	13-10-2026 02:56 PM	Java Source File	1 KB
Ticket_Reservation_System.java	13-10-2026 02:56 PM	Java Source File	1 KB
Train.java	13-10-2026 02:56 PM	Java Source File	1 KB

```java
public class Admin {
    /**
     * Default constructor
     */
    public Admin() {
    }
    /**
     *
     */
    private int adminId;
    /**
     *
     */
    private string name;
    /**
     *
     */
    private string role;
    /**
     * @return
     */
    public void manageSchedule() {
        // TODO implement here
        return null;
    }
    /**
     * @return
     */
    public void manageBooking() {
        // TODO implement here
        return null;
    }
    /**
     * @return
     */
    public void generateReport() {
        // TODO implement here
        return null;
    }
}
```

```java
public class Booking {
    /**
     * Default constructor
     */
    public Booking() {
    }
    /**
     *
     */
    private int bookingId;
    /**
     *
     */
    private date bookingDate;
    /**
     *
     */
    private string bookingStatus;
    /**
     *
     */
    public void Port1;
    /**
     *
     */
    public void createBooking() {
        // TODO implement here
    }
    /**
     *
     */
    public void cancelBooking() {
        // TODO implement here
    }
    /**
     *
     */
    public void modifyBooking() {
        // TODO implement here
    }
}
```

```java
public class Payment {

    /**
     * Default constructor
     */
    public Payment() {
    }

    /**
     *
     */
    private int paymentId;

    /**
     *
     */
    private double amount;

    /**
     *
     */
    private string paymentStatus;
    /**
     *
     */
    public void processPayment() {
        // TODO implement here
    }
    /**
     *
     */
    public void verifyPayment() {
        // TODO implement here
    }

}
```

```java
public class Ticket_Reservation_System {

    /**
     * Default constructor
     */
    public Ticket_Reservation_System() {
    }

    /**
     *
     */
    private int systemId;

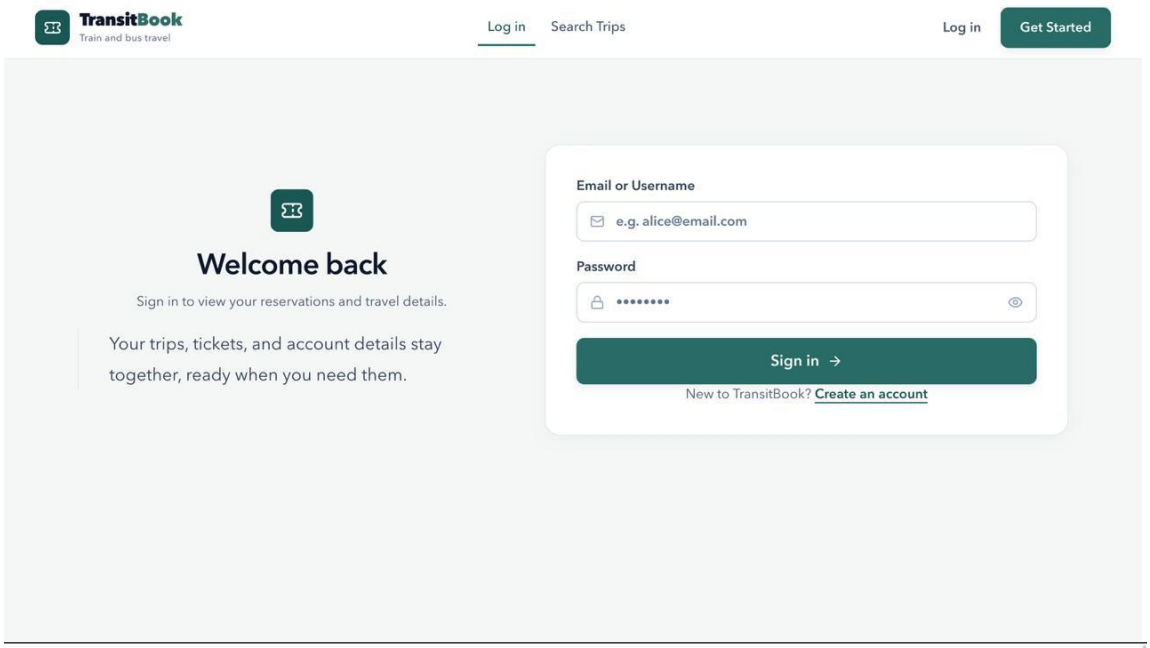
    /**
     *
     */
    private string systemName;
    /**
     *
     */
    public void manageBooking() {
        // TODO implement here
    }

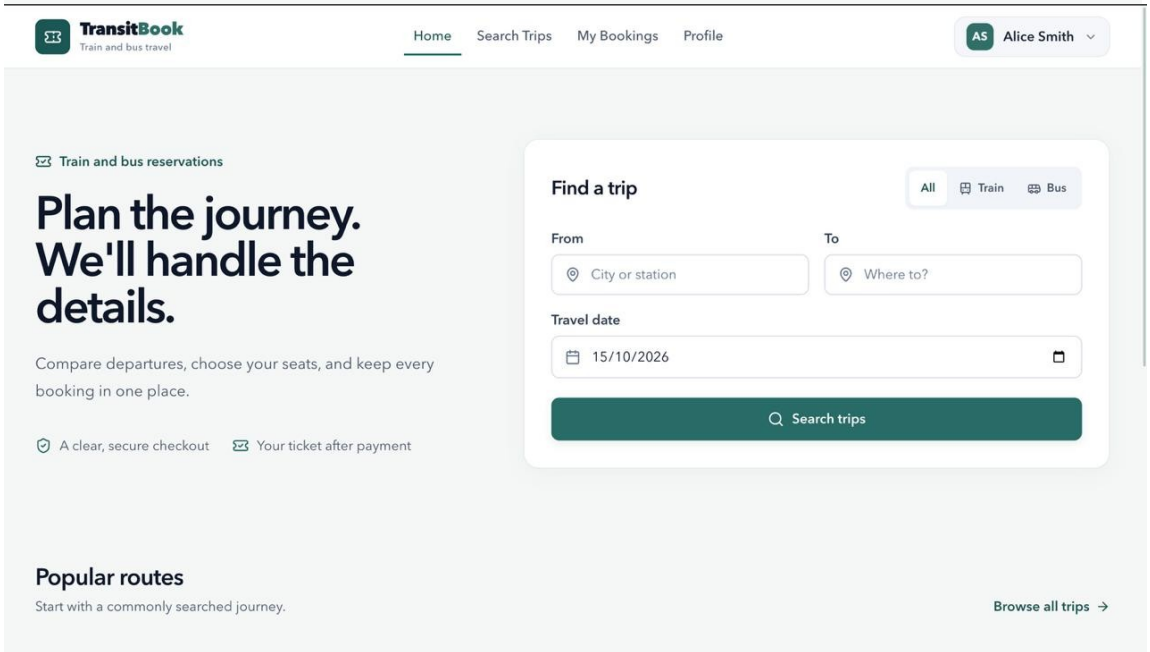
    /**
     *
     */
    public void managePassenger() {
        // TODO implement here
    }

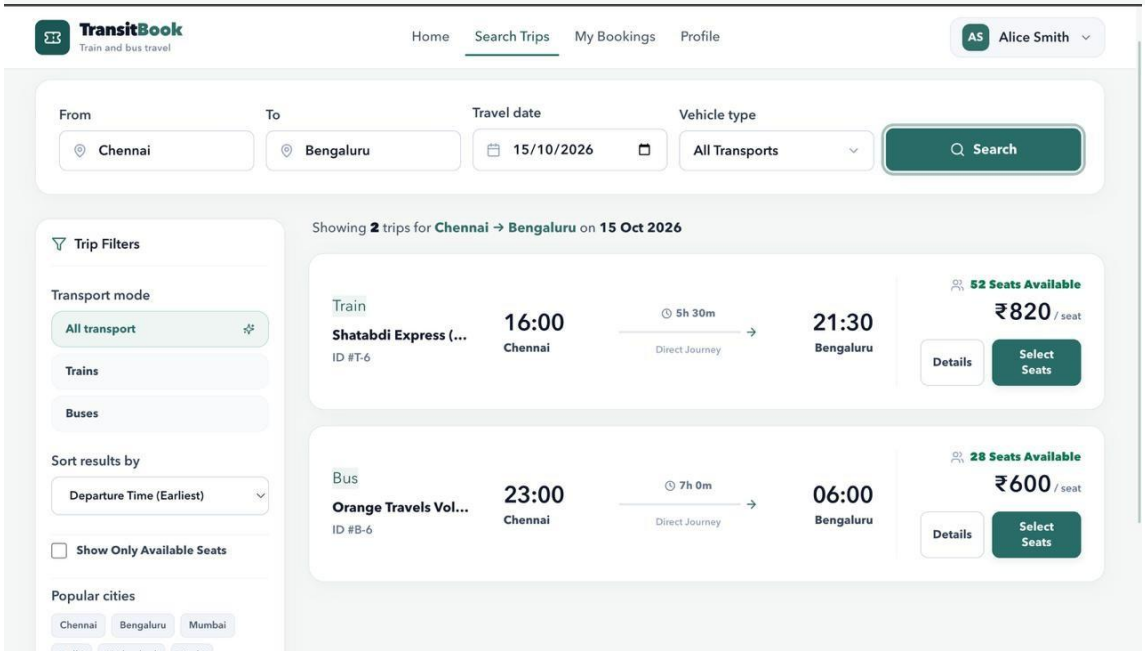
    /**
     *
     */
    public void generateReport() {
        // TODO implement here
    }

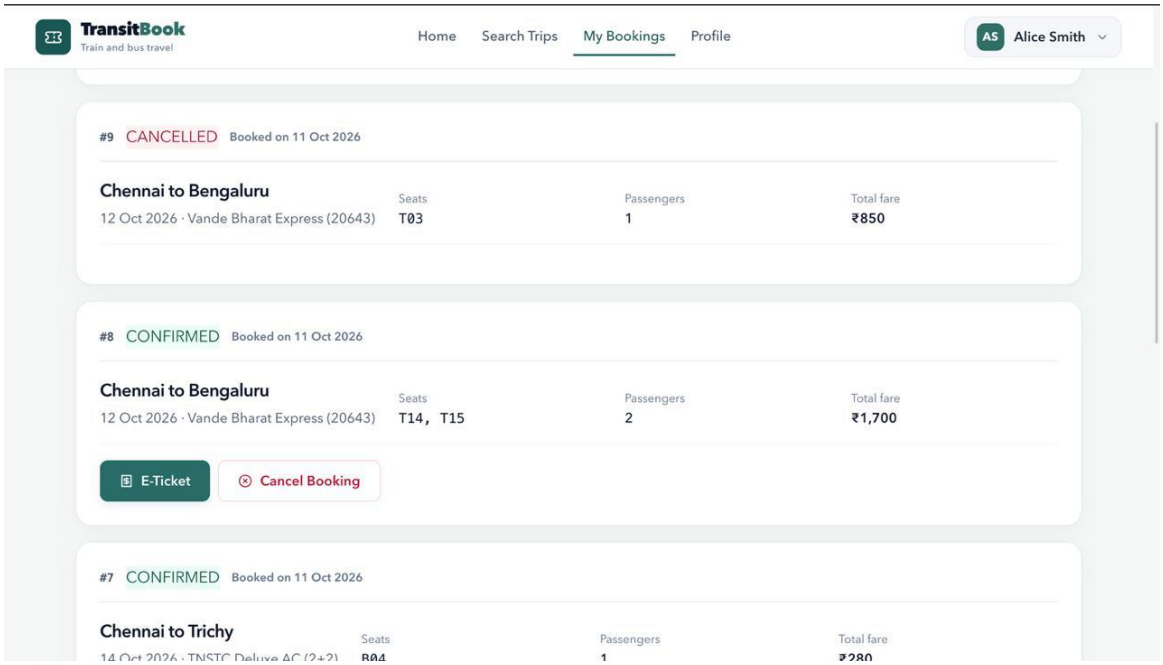
}
```

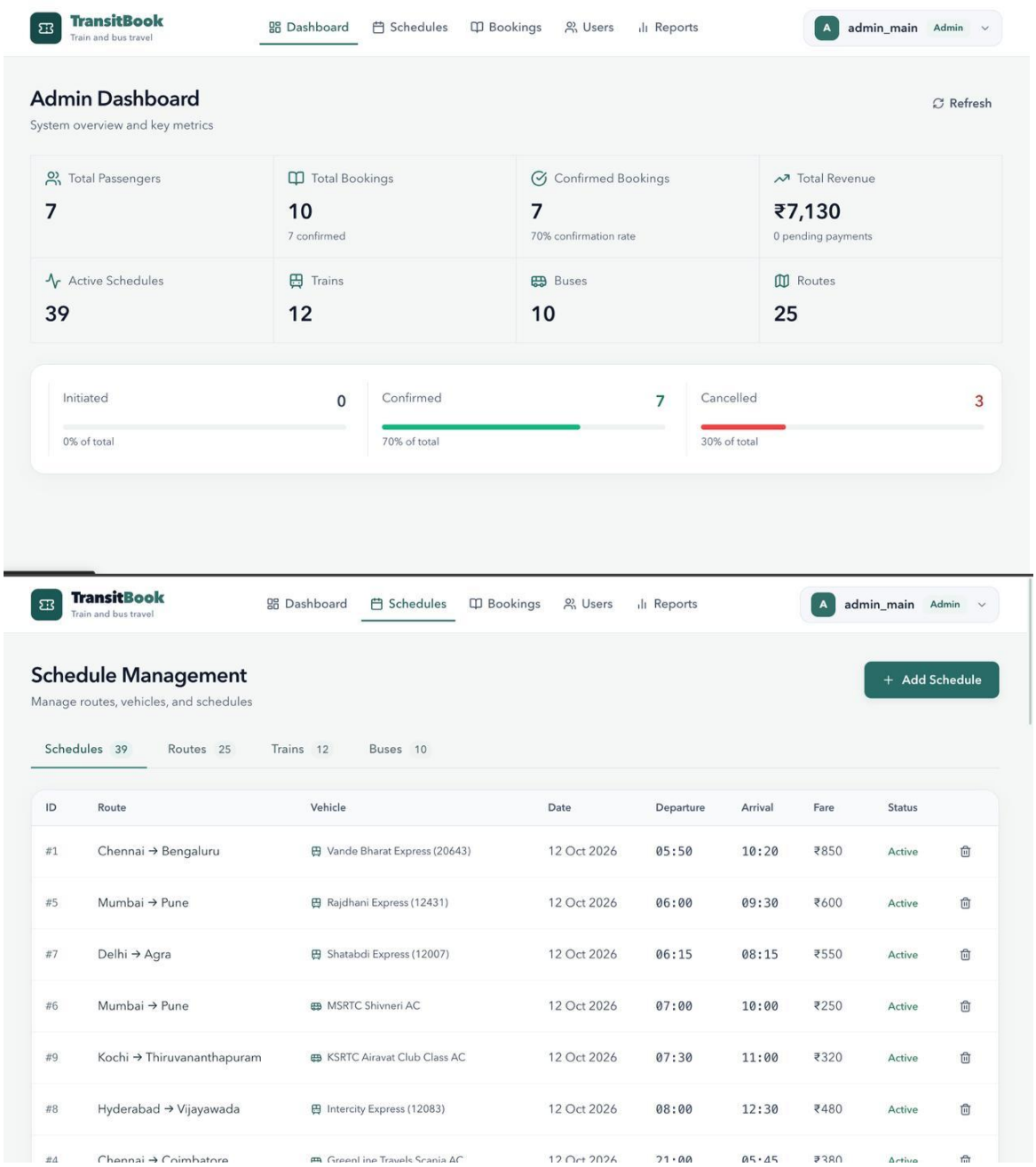
Sample UI Design:








TransitBook
Train and bus travel
Home
Search Trips
My Bookings
Profile
AS Alice Smith
#9 CANCELLED Booked on 11 Oct 2026
Chennai to Bengaluru
12 Oct 2026 · Vande Bharat Express (20643)
Seats
T03
Passengers
1
Total fare
₹850
#8 CONFIRMED Booked on 11 Oct 2026
Chennai to Bengaluru
12 Oct 2026 · Vande Bharat Express (20643)
Seats
T14, T15
Passengers
2
Total fare
₹1,700
E-Ticket
Cancel Booking
#7 CONFIRMED Booked on 11 Oct 2026
Chennai to Trichy
Seats
B04
Passengers
1
Total fare
₹280


TransitBook
Train and bus travel
Dashboard
Schedules
Bookings
Users
Reports
A admin_main Admin
Admin Dashboard
System overview and key metrics
Refresh
Total Passengers
7
Total Bookings
10
7 confirmed
Confirmed Bookings
7
70% confirmation rate
Total Revenue
₹7,130
0 pending payments
Active Schedules
39
Trains
12
Buses
10
Routes
25
Initiated
0
0% of total
Confirmed
7
70% of total
Cancelled
3
30% of total

TransitBook
Train and bus travel
Dashboard
Schedules
Bookings
Users
Reports
A admin_main Admin
Schedule Management
Manage routes, vehicles, and schedules
+ Add Schedule
Schedules 39
Routes 25
Trains 12
Buses 10
ID
Route
Vehicle
Date
Departure
Arrival
Fare
Status
#1 Chennai → Bengaluru Vande Bharat Express (20643) 12 Oct 2026 05:50 10:20 ₹850 Active
#5 Mumbai → Pune Rajdhani Express (12431) 12 Oct 2026 06:00 09:30 ₹600 Active
#7 Delhi → Agra Shatabdi Express (12007) 12 Oct 2026 06:15 08:15 ₹550 Active
#6 Mumbai → Pune MSRTC Shivneri AC 12 Oct 2026 07:00 10:00 ₹250 Active
#9 Kochi → Thiruvananthapuram KSRTC Airavat Club Class AC 12 Oct 2026 07:30 11:00 ₹320 Active
#8 Hyderabad → Vijayawada Intercity Express (12083) 12 Oct 2026 08:00 12:30 ₹480 Active
#4 Chennai → Coimbatore GreenLine Travels Scania AC 12 Oct 2026 21:00 05:45 ₹380 Active

TransitBook
Train and bus travel

Dashboard | Schedules | Bookings | Users | Reports

admin_main Admin

Bookings Overview

10 total bookings

Refresh

Search passenger, route, booking ID...

Booking ID	Passenger	Route	Type	Date	Seats	Amount	Status
#10	Alice Smith alice@email.com	Chennai → Bengaluru	Train	15 Oct 2026	1 seat(s)	₹820	CANCELLED
#9	Alice Smith alice@email.com	Chennai → Bengaluru	Train	12 Oct 2026	1 seat(s)	Not available	CANCELLED
#8	Alice Smith alice@email.com	Chennai → Bengaluru	Train	12 Oct 2026	2 seat(s)	₹1,700	CONFIRMED
#7	Alice Smith alice@email.com	Chennai → Trichy	Bus	14 Oct 2026	1 seat(s)	₹280	CONFIRMED
#6	Alice Smith alice@email.com	Chennai → Salem	Bus	15 Oct 2026	2 seat(s)	₹500	CONFIRMED
#5	Alice Smith alice@email.com	Chennai → Coimbatore	Train	16 Oct 2026	2 seat(s)	₹880	CONFIRMED

TransitBook
Train and bus travel

Dashboard | Schedules | Bookings | Users | Reports

admin_main Admin

User Management

7 passengers · 0 admins

Refresh

Search username, email, or name...

User	Email	Role	Passenger Info	Member Since	Status	Actions
R rahul_v	rahul@email.com	PASSENGER	9876543216	11 Oct 2026	Active	Change
P priya_r	priya@email.com	PASSENGER	9876543215	11 Oct 2026	Active	Change
E eva_green	eva@email.com	PASSENGER	9876543214	11 Oct 2026	Active	Change
D david_m	david@email.com	PASSENGER	9876543213	11 Oct 2026	Active	Change
C charlie_k	charlie@email.com	PASSENGER	9876543212	11 Oct 2026	Active	Change
B bob_runner	bob@email.com	PASSENGER	9876543211	11 Oct 2026	Active	Change

TransitBook
Train and bus travel

Dashboard | Schedules | Bookings | Users | Reports

admin_main Admin

Reports and analytics

Revenue, booking, passenger, and route performance.

Refresh

Revenue

Total revenue

₹7,130

By payment method

Card	₹2,580 (2 payments)
UPI	₹4,550 (6 payments)

Bookings

10	7	0	3
Total	Confirmed	Initiated	Cancelled

Passengers